

A SHARP–DIFFUSE INTERFACE MODEL FOR INTERMITTENT AND ISOLATED TOPOLOGICAL TRANSITIONS

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Abstract. We propose a hybrid sharp–diffuse interface representation for modeling intermittent and isolated topological transitions during Cahn–Hilliard phase coarsening. Away from topological events, the evolution is approximated by the Mullins–Sekerka sharp-interface limit system and computed using a boundary integral formulation. When diffuse transition layers overlap, a novel *interface surgery* algorithm resolves topology changes through a localized Cahn–Hilliard pseudo-time evolution, after which the sharp-interface calculation is resumed. We develop the mathematical formulation underlying this decomposition, present a simple two-dimensional numerical implementation, and simulate a mass-exchange problem with interface coalescence. By localizing the diffuse evolution to topological events, the method achieves a speedup of two to three orders of magnitude over conventional diffuse-interface simulations.

Key words. Cahn–Hilliard, Mullins–Sekerka, topological transition, boundary integral

MSC codes. 76T99, 76M15, 35Q35

1. Introduction. The description of an interface as a thin transition layer separating two distinct phases traces back to the foundational work of van der Waals and Korteweg. Building on this diffuse-interface picture, Cahn and Hilliard [5] developed a variational model for phase separation and coarsening in binary alloys subjected to rapid cooling, or quenching. The quench renders an initially homogeneous mixture thermodynamically unstable: it quickly *separates* into nearly pure phases divided by thin transition layers, and then *coarsens* more slowly as diffusion increases the characteristic size of these regions. The *Cahn–Hilliard equation* (see (2.2) below), which is the fundamental diffuse-interface model studied in this work, represents this two-stage evolution as a mass-conserving gradient flow of the Ginzburg–Landau free energy. This energetic framework is highly flexible and has been applied across a wide range of disciplines [3, 6, 22, 32].

The Cahn–Hilliard equation and related systems have also been studied extensively in the mathematical literature. For a comprehensive treatment, we refer to the monograph of Miranville [28]. Elliott and Zheng [18] established existence, uniqueness, and regularity for the constant-mobility case with a polynomial free-energy density, while Nicolaenko–Scheurer–Temam [31] analyzed the long-time behavior. In the phase-coarsening regime of interest here, heuristic scaling predicts that the coarsening length scale $\ell(t)$, measuring the typical size of the phase domains, grows according to the slow law $\ell(t) \sim t^{1/3}$ [23].

During phase coarsening, interfaces can undergo topology changes, which we assume are intermittent and isolated in a sense that will be made precise below. These topological events are governed by diffuse mixing on spatial and temporal scales of order $\mathcal{O}(\varepsilon)$ and $\mathcal{O}(\varepsilon^3)$, respectively, where ε denotes the transition-layer thickness. In practical applications, ε is much smaller than the macroscopic length scale L of the interfacial evolution. Direct diffuse-interface simulations must resolve both the transition layers and the topological events with sufficient spatiotemporal resolution. High-order spatial discretizations [26], energy-stable semi-implicit time integrators [37, 38], and sophisticated adaptive mesh-refinement strategies [41] have enabled sim-

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ulations of complex multiscale phenomena, but entail significant computational cost and implementation complexity.

These computational difficulties motivate the study of the sharp-interface limit $\varepsilon \rightarrow 0$ of the Cahn–Hilliard system. The limiting equations, the Mullins–Sekerka system (see (3.20) below), were formally derived by Pego [33]. The convergence of the Cahn–Hilliard system to the Mullins–Sekerka system was subsequently justified rigorously by Alikakos–Bates–Chen [2], provided that the limiting system admits a smooth solution. Local well-posedness and instantaneous regularization of the limiting evolution were established by Chen–Hong–Yi [9]. For a broader discussion of related sharp-interface limits, we refer the reader to the recent reviews of Abels and Garcke [1] and Du and Feng [15].

The mathematical theory justifying the sharp-interface limit breaks down when diffuse transition layers overlap, thereby excluding topological transitions such as coalescence, a fundamental feature of phase coarsening. Numerical solvers based on sharp-interface formulations encounter the same difficulty; the method of Zhu–Chen–Hou [42], for example, necessarily terminates prior to a topology change. Moreover, the authors explicitly note that extensions to more complex phenomena and larger-scale problems first require a robust method for resolving topological singularities.

In this work, we introduce a novel mathematical and numerical model, based on a hybrid sharp–diffuse formulation, for continuing the evolution of an interface through *intermittent* and *isolated* topological transitions arising from Cahn–Hilliard dynamics.¹ We formulate this approach in three principal steps:

- approximation of the diffuse solution during “regular” evolution by its sharp-interface limit, together with an equivalent boundary-integral formulation;
- geometric characterization of the sharp-interface regime, with its breakdown triggering the suspension of the sharp-interface evolution;
- and, finally, resolution of the topological transition through a localized diffuse relaxation, followed by restart of the sharp-interface evolution.

Numerically, these three steps are realized using established computational tools: the θ – L representation of the evolving interface [42], the manifold-death algorithm [10], and a smooth auxiliary variable (SAV) scheme [37], respectively.

More generally, we note that many other numerical techniques have been developed to handle topological changes in fluid flows, including Chorin’s vortex-filament reconnection algorithm [11], the front-tracking methods of Glimm et al. [20] and Tryggvason et al. [40], and specialized procedures for boundary-integral simulations [14, 43]. To the best of our knowledge, however, all such methods resolve the topology change by numerical regularization, often through an ad hoc rule. Lowengrub and Truskinovsky [27] have demonstrated that the interfacial stresses are sensitive to the choice of regularization when geometric length scales become comparable to the interfacial thickness, as occurs near a topology change. Our sharp–diffuse model instead resolves the topological transition through physical diffuse-interface dynamics.

To illustrate numerically the proposed methodology, we consider a four-circle coalescence problem originally studied by Zhu–Chen–Hou [42]. Whereas their sharp-interface calculation terminated before coalescence, our method resolves the topology change and continues the evolution beyond it. Moreover, benchmarks against an established energy-stable pseudospectral solver for the fully diffuse system show speedups of two to three *orders of magnitude*. We refer the reader to Figures 5 and 6

¹To our knowledge, the underlying hybrid viewpoint was first suggested by Lowengrub and Truskinovsky [27].

below for a depiction of the complete evolution—from the breakdown of the sharp-interface regime, through localized diffuse relaxation, to the restarted sharp-interface evolution—realized by our sharp–diffuse numerical model.

Outline of the paper. The remainder of the paper is structured as follows. Section 2 introduces the diffuse-interface model and recalls its basic properties. Section 3 describes the basic geometry for the formal passage to the sharp-interface limit, recalls the known mathematical results justifying the sharp-interface approximation, and presents an equivalent boundary-integral formulation of the limiting evolution. The main contributions appear in Sections 4 to 6, which formulate the sharp–diffuse continuation procedure described above. We summarize our simple two-dimensional numerical implementation in Section 7 and test the method on the four-circle coalescence problem in Section 8. Finally, we provide concluding remarks in Section 9. Appendix A derives an explicit polynomial profile for idealized coalescence, while the supplementary material provides a video of the four-circle simulation produced using our sharp–diffuse method.

2. Diffuse-interface model. Consider a binary mixture occupying $\mathbb{R}^d$, for $d = 2, 3$, with its local composition described by a *phase-field function*

$$(2.1) \quad \phi_\varepsilon : \mathbb{R}^d \times \mathbb{R}^+ \rightarrow [-1, 1].$$

The values ± 1 represent the two pure phases, which are assumed to be separated by a diffuse transition layer of thickness $\mathcal{O}(\varepsilon)$. A schematic of the setup is shown in Figure 1. For each time $t > 0$, the mixture evolves according to

$$(2.2a) \quad \partial_t \phi_\varepsilon(x, t) = m_\varepsilon \Delta \mu_\varepsilon(x, t),$$

$$(2.2b) \quad \mu_\varepsilon(x, t) = \hat{\sigma} \left(-\varepsilon \Delta \phi_\varepsilon(x, t) + \frac{1}{\varepsilon} W'(\phi_\varepsilon) \right).$$

Here $\mu_\varepsilon(x, t)$ is the chemical potential, which measures the first-order change in free energy produced by a local change in composition; spatial gradients of μ_ε drive diffusive transport, with the mobility parameter $m_\varepsilon > 0$ controlling the rate of this transport. The constant $\hat{\sigma} > 0$ is the diffuse capillary-energy coefficient.

The function W is a smooth, nonnegative double-well potential satisfying

$$(2.3a) \quad W(\pm 1) = W'(\pm 1) = 0, \quad W''(\pm 1) > 0.$$

In this work, we consider the standard quartic polynomial approximation

$$(2.3b) \quad W(\phi) = \frac{1}{4}(1 - \phi^2)^2.$$

The system (2.2) is initialized with

$$(2.4) \quad \phi_\varepsilon(x, 0) = \mathring{\phi}_\varepsilon(x), \quad x \in \mathbb{R}^d,$$

where $\mathring{\phi}_\varepsilon(x)$ is the prescribed initial datum. No initial condition is imposed on $\mu_\varepsilon(x, 0)$, which is determined from $\phi_\varepsilon(x, 0)$ by (2.2b).

We consider the equations (2.2) posed on $\mathbb{R}^d$ with the far-field conditions

$$(2.5) \quad \phi_\varepsilon(x, t) \longrightarrow +1, \quad \mu_\varepsilon(x, t) \longrightarrow 0 \quad \text{sufficiently rapidly as } |x| \longrightarrow \infty.$$

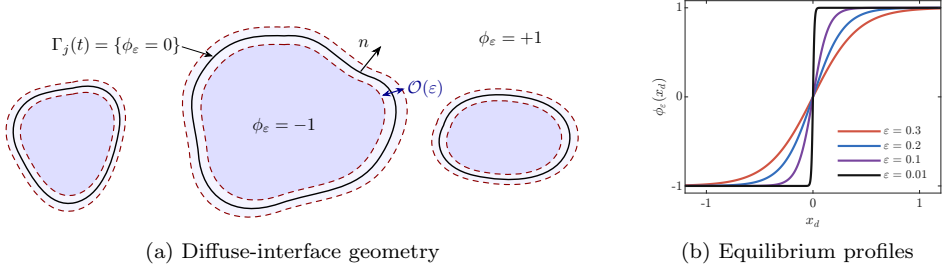


FIG. 1. **Left:** Schematic of the phase-field problem. The solid curves represent the level sets $\phi_\varepsilon = 0$, while the dashed curves mark transition layers of thickness $\mathcal{O}(\varepsilon)$ separating the two pure phases $\phi = \pm 1$. **Right:** Equilibrium profile $q(x_d/\varepsilon)$, defined in (2.10), for several values of ε .

2.1. Free energy and chemical potential. The system (2.2) is the *Cahn–Hilliard system*, derived by Cahn and Hilliard [5] to model phase separation and coarsening in binary mixtures. In this model, the interfacial energy carried by a finite-width transition layer is described by the Ginzburg–Landau free energy,

$$(2.6) \quad E_\varepsilon[\phi] = \hat{\sigma} \int_{\mathbb{R}^d} \left(\frac{\varepsilon}{2} |\nabla \phi|^2 + \frac{1}{\varepsilon} W(\phi) \right) dx.$$

The potential term causes the system to favor configurations containing the two pure phases $\phi = \pm 1$, while the gradient term penalizes rapid spatial variation of the phase field. The two terms are weighted according to the Modica–Mortola scaling [30], producing diffuse transition layers of thickness $\mathcal{O}(\varepsilon)$ between the pure phases (see Lemma 2.1 below).

The chemical potential describes the first-order change in free energy produced by a local change in ϕ , and thereby determines the energetic driving force for diffusion. More precisely, for a smooth compactly supported test function $\psi \in C_c^\infty(\mathbb{R}^d)$, the first variation of (2.6) in the direction ψ is

$$DE_\varepsilon[\phi](\psi) := \frac{d}{d\delta} E_\varepsilon[\phi + \delta\psi] \Big|_{\delta=0} = \hat{\sigma} \int_{\mathbb{R}^d} \left(-\varepsilon \Delta \phi + \frac{1}{\varepsilon} W'(\phi) \right) \psi dx,$$

and since ψ is arbitrary, it follows from (2.2b) that

$$(2.7) \quad \mu_\varepsilon = \frac{\delta E_\varepsilon}{\delta \phi}[\phi_\varepsilon].$$

2.2. Equilibrium transition profile. The structure of a single flat diffuse interface can be determined directly from the free energy. Consider a stationary layer centered on the plane $\{x_d = 0\}$, and write

$$(2.8) \quad \phi_\varepsilon(x) = q\left(\frac{x_d}{\varepsilon}\right), \quad \lim_{z \rightarrow -\infty} q(z) = -1, \quad \lim_{z \rightarrow +\infty} q(z) = +1.$$

At equilibrium, the first variation of the free energy vanishes, $\mu_\varepsilon = 0$. Substitution of (2.8) into (2.2b) then gives the one-dimensional equilibrium equation

$$(2.9) \quad -q'' + W'(q) = 0, \quad q(-\infty) = -1, \quad q(+\infty) = +1.$$

If $q(z)$ solves (2.9), then so does $q(z - z_0)$ for any $z_0 \in \mathbb{R}$. We center the transition layer at $z = 0$ by imposing $q(0) = 0$. The resulting profile and the associated interfacial energy may be computed explicitly.

LEMMA 2.1 (Equilibrium transition profile). *For the potential (2.3b), the unique increasing solution of (2.9) satisfying $q(0) = 0$ is*

$$(2.10) \quad q(z) = \tanh\left(\frac{z}{\sqrt{2}}\right).$$

The associated interfacial energy is

$$(2.11) \quad \sigma := \widehat{\sigma} c_W, \quad c_W := \int_{-\infty}^{\infty} |q'(z)|^2 dz = \int_{-1}^1 \sqrt{2W(s)} ds = \frac{2\sqrt{2}}{3}.$$

Plots of the profile (2.10) are shown in Figure 1b for different values of ε .

2.3. Diffuse transition-layer thickness. The equilibrium profile (2.10) provides a natural geometric description of the diffuse interface. Let $\Gamma \subset \mathbb{R}^d$ be a smooth closed interface enclosing a bounded region D , and define its signed distance by

$$(2.12) \quad d_\Gamma(x) := \begin{cases} -\text{dist}(x, \Gamma), & x \in D, \\ \text{dist}(x, \Gamma), & x \in \mathbb{R}^d \setminus \overline{D}, \\ 0, & x \in \Gamma. \end{cases}$$

The diffuse field associated with Γ is obtained by placing the one-dimensional profile (2.10) across the interface:

$$(2.13) \quad \phi_\varepsilon^\Gamma(x) := q\left(\frac{d_\Gamma(x)}{\varepsilon}\right) = \tanh\left(\frac{d_\Gamma(x)}{\varepsilon\sqrt{2}}\right).$$

Since (2.10) approaches ± 1 only asymptotically, a finite transition-layer width must be specified by fixing a pair of level sets. For example, we can define an effective transition layer as the region between the ± 0.9 level sets of $\phi_\varepsilon^\Gamma(x)$. Using (2.13), this region can be written equivalently in terms of $d_\Gamma(x)$ as

$$(2.14a) \quad \mathcal{T}_\varepsilon[\Gamma] := \{x \in \mathbb{R}^d : |\phi_\varepsilon^\Gamma(x)| < 0.9\} = \{x \in \mathbb{R}^d : |d_\Gamma(x)| < C_{\text{layer}}\varepsilon\},$$

$$(2.14b) \quad C_{\text{layer}} = \sqrt{2} \operatorname{arctanh}(0.9) \approx 2,$$

so that $C_{\text{layer}}\varepsilon \approx 2\varepsilon$ is the effective half-width of the diffuse layer.

2.4. Energy dissipation and mass conservation. Equation (2.2a) may be written in flux form with diffusive flux f_ε ,

$$(2.15a) \quad \partial_t \phi_\varepsilon(x, t) + \nabla \cdot f_\varepsilon(x, t) = 0,$$

$$(2.15b) \quad f_\varepsilon(x, t) = -m_\varepsilon \nabla \mu_\varepsilon(x, t).$$

The first equation expresses local conservation of composition, while the second states that the diffusive flux is directed down gradients of the chemical potential. For the closed-interface configurations considered here, we take $\phi_\varepsilon = +1$ to be the exterior phase. The mass of the -1 phase is then defined by

$$(2.16) \quad \mathcal{M}_\varepsilon(t) := \int_{\mathbb{R}^d} \frac{1}{2} (1 - \phi_\varepsilon(x, t)) dx,$$

provided $1 - \phi_\varepsilon(\cdot, t) \in L^1(\mathbb{R}^d)$.

LEMMA 2.2 (Energy dissipation and mass conservation). *Let $(\phi_\varepsilon, \mu_\varepsilon)$ be a sufficiently regular solution of (2.2) satisfying the stated decay assumptions, and suppose that $1 - \phi_\varepsilon(\cdot, t) \in L^1(\mathbb{R}^d)$. Then*

$$(2.17) \quad \frac{d}{dt} E_\varepsilon[\phi_\varepsilon] = -m_\varepsilon \int_{\mathbb{R}^d} |\nabla \mu_\varepsilon|^2 dx \leq 0.$$

Moreover, the phase mass (2.16) is conserved.

2.5. The H^{-1} gradient-flow structure. The first variation of the free energy identifies the chemical potential, but it does not by itself determine the evolution. The evolution is obtained only after specifying the metric in which the free energy decreases most rapidly. Because the gradient flow is restricted to phase fields with fixed phase mass, every admissible variation ψ must satisfy $\int_{\mathbb{R}^d} \psi dx = 0$. We denote the homogeneous H^{-1} inner product on smooth, decaying, zero-mean functions by

$$(2.18) \quad \langle u, v \rangle_{H^{-1}} := \int_{\mathbb{R}^d} (-\Delta)^{-1} u v dx.$$

LEMMA 2.3 (H^{-1} -gradient-flow structure). *Let ϕ_ε be a smooth phase field satisfying the stated decay assumptions, and let μ_ε be the chemical potential defined by (2.7). Then $\text{grad}_{H^{-1}} E_\varepsilon[\phi_\varepsilon] = -\Delta \mu_\varepsilon$, so that (2.2a) may be written as*

$$(2.19a) \quad \partial_t \phi_\varepsilon = -m_\varepsilon \text{grad}_{H^{-1}} E_\varepsilon[\phi_\varepsilon],$$

and the corresponding gradient-flow dissipation identity is

$$(2.19b) \quad \frac{d}{dt} E_\varepsilon[\phi_\varepsilon] = -m_\varepsilon \|\text{grad}_{H^{-1}} E_\varepsilon[\phi_\varepsilon]\|_{H^{-1}}^2.$$

3. Sharp-interface limit. We are interested in the sharp-interface limit of (2.2) as $\varepsilon \rightarrow 0$, corresponding to the physical regime in which the width of the diffuse transition layer is negligible relative to the geometric length scales of the phase domains. We assume that the mobility remains nondegenerate in this limit:

$$(3.1) \quad m_\varepsilon = m_0 + \mathcal{O}(\varepsilon), \quad m_0 > 0,$$

so that diffusion remains active in the bulk phases and disconnected components can exchange mass through the surrounding mixture.

3.1. Sharp-interface geometry. Fix $T > 0$ and consider N pairwise disjoint closed interfaces $\Gamma_j(t)$ evolving for $t \in [0, T]$, with $D_j^-(t)$ denoting the bounded domain enclosed by $\Gamma_j(t)$. We write

$$(3.2a) \quad \Gamma(t) := \bigcup_{j=1}^N \Gamma_j(t), \quad D^-(t) := \bigcup_{j=1}^N D_j^-(t), \quad D^+(t) := \mathbb{R}^d \setminus \overline{D^-(t)}$$

We assume that each interface component admits a parametrization

$$(3.2b) \quad X_j : \mathbb{S}^{d-1} \times [0, T] \longrightarrow \mathbb{R}^d, \quad \Gamma_j(t) = X_j(\mathbb{S}^{d-1}, t),$$

where $X_j(\cdot, t)$ is an embedding for every $t \in [0, T]$ and

$$(3.2c) \quad X_j \in C^\infty(\mathbb{S}^{d-1} \times [0, T]; \mathbb{R}^d).$$

Thus, each $\Gamma_j(t)$ is a smooth closed hypersurface.

The total interfacial measure is denoted by

$$(3.3) \quad |\mathbf{\Gamma}(t)| := \mathcal{H}^{d-1}(\mathbf{\Gamma}(t)) = \sum_{j=1}^N \mathcal{H}^{d-1}(\Gamma_j(t)),$$

where $\mathcal{H}^{d-1}$ denotes the $(d-1)$ -dimensional Hausdorff measure. Thus, (3.3) gives the total interfacial length when $d = 2$ and the total interfacial area when $d = 3$.

The limiting phase field $\phi_0(x, t)$ is assumed to take the form

$$(3.4) \quad \phi_0(x, t) = \begin{cases} -1, & x \in D^-(t), \\ +1, & x \in D^+(t). \end{cases}$$

Equivalently, $\phi_0(\cdot, t) = 1 - 2\chi_{D^-(t)}$, where χ_A denotes the indicator function on the set A . With a slight abuse of notation, we denote the time-dependent signed-distance function and transition layers using the same notation in (2.12) and (2.14),

$$d_{\mathbf{\Gamma}}(x, t) := d_{\mathbf{\Gamma}(t)}(x), \quad \mathcal{T}_{\varepsilon}(t) := \mathcal{T}_{\varepsilon}[\mathbf{\Gamma}(t)].$$

3.1.1. Reach and signed-distance coordinates. The geometry of $\mathbf{\Gamma}(t)$ can be described using smooth signed-distance (or normal) coordinates wherever each nearby point has a unique nearest point on the interface. The size of this neighborhood is quantified by the following definition.

DEFINITION 3.1 (Reach [19]). *The reach of a closed set $A \subset \mathbb{R}^d$ is defined as*

$$(3.5) \quad \text{reach}(A) := \sup \left\{ \rho > 0 : \begin{array}{l} \text{every } x \in \mathbb{R}^d \text{ satisfying } \text{dist}(x, A) < \rho \\ \text{has a unique nearest point in } A \end{array} \right\}.$$

A minimal geometric requirement for the sharp-interface approximation to remain valid is that the associated diffuse transition layers do not overlap; see Figure 2. The separation condition can be characterized directly in terms of the reach, leading to the following basic assumption on the sharp-interface evolution.

ASSUMPTION 3.2 (Separated transition layers). *The transition layers remain well separated on $[0, T]$, in the sense that*

$$(3.6) \quad \text{reach}(\mathbf{\Gamma}(t)) > C_{\text{layer}}\varepsilon \quad \text{for every } t \in [0, T],$$

where C_{layer} is defined in (2.14).

On any interval $[0, T]$ on which Assumption 3.2 holds, define

$$(3.7) \quad \rho_* := \inf_{t \in [0, T]} \text{reach}(\mathbf{\Gamma}(t)) \geq C_{\text{layer}}\varepsilon.$$

For any $0 < \rho < \rho_*$, we define a tubular neighborhood of $\mathbf{\Gamma}(t)$ by

$$(3.8) \quad \mathcal{N}_{\rho}(t) := \{x \in \mathbb{R}^d : \text{dist}(x, \mathbf{\Gamma}(t)) < \rho\}, \quad \mathcal{N}_{\rho} := \bigcup_{0 \leq t \leq T} \mathcal{N}_{\rho}(t) \times \{t\},$$

where $\mathcal{N}_{\rho} \subset \mathbb{R}^d \times [0, T]$ denotes the spacetime tube swept out by $\mathcal{N}_{\rho}(t)$.

The assumed smoothness of the parametrization (3.2c) and the uniform reach bound (3.6) yield the following normal-coordinate representation.²

²The fixed-time statement follows from [16, Lemma 2.8], while the smooth dependence on time follows from (3.2c).

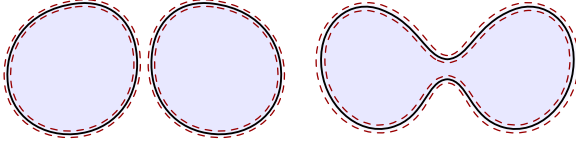


FIG. 2. **Configuration satisfying the reach condition** (3.6). The solid black curves represent the sharp interfaces, while the dashed red curves delimit diffuse transition layers of effective half-width $C_{\text{layer}}\varepsilon$. Although the reach is small, the transition layers remain disjoint in both configurations. A further decrease in reach would cause the layers to overlap, signaling an impending coalescence of distinct components (left) or pinch-off of a single component (right).

LEMMA 3.3 (Signed-distance coordinates). *For every $0 < \rho < \rho_*$ and every $(x, t) \in \mathcal{N}_\rho$, there is a unique nearest point $\pi(x, t) \in \Gamma(t)$. Moreover,*

$$(3.9) \quad d_\Gamma \in C^\infty(\mathcal{N}_\rho), \quad \pi \in C^\infty(\mathcal{N}_\rho; \mathbb{R}^d),$$

and each $x \in \mathcal{N}_\rho(t)$ admits the representation

$$(3.10) \quad x = \pi(x, t) + d_\Gamma(x, t) \nabla d_\Gamma(x, t), \quad |\nabla d_\Gamma(x, t)| = 1.$$

3.1.2. Normal and curvature conventions. Let $\alpha = (\alpha_1, \dots, \alpha_{d-1})$ denote local coordinates on $\mathbb{S}^{d-1}$. Because $X_j(\cdot, t)$ is a smooth embedding, we may define a smooth unit normal on $\Gamma_j(t)$ by

$$(3.11a) \quad n(X_j(\alpha, t), t) := \begin{cases} \frac{(\partial_\alpha X_j)^\perp}{|\partial_\alpha X_j|}, & d = 2, \\ \frac{\partial_{\alpha_1} X_j \times \partial_{\alpha_2} X_j}{|\partial_{\alpha_1} X_j \times \partial_{\alpha_2} X_j|}, & d = 3, \end{cases}$$

where $v^\perp = (-v_2, v_1)$ and the local coordinates are oriented so that n points from the -1 phase into the $+1$ phase. By Lemma 3.3, the signed-distance function is smooth throughout $\mathcal{N}_\rho$, and therefore defines the canonical extension of the interface normal

$$(3.11b) \quad n(x, t) := \nabla d_\Gamma(x, t) = n(\pi(x, t), t), \quad (x, t) \in \mathcal{N}_\rho.$$

It then follows from (3.10) that every point in $\mathcal{N}_\rho$ can be represented as

$$(3.11c) \quad x = \pi(x, t) + d_\Gamma(x, t) n(x, t).$$

For $(x, t) \in \mathcal{N}_\rho$, define the tangential projection $P_\tau := I - n \otimes n$. At each $x \in \mathcal{N}_\rho(t)$, P_τ projects orthogonally onto the tangent space of the signed-distance level set through x ; when $x \in \Gamma(t)$, this space is precisely $T_x \Gamma(t)$. For a scalar function f and a vector field v , we define the normal derivative, tangential gradient, and tangential divergence by

$$\partial_n f := n \cdot \nabla f, \quad \partial_\tau f := P_\tau \nabla f, \quad \partial_\tau \cdot v := \text{tr}(P_\tau \nabla v).$$

The scalar mean curvature is defined by

$$(3.12a) \quad \kappa := \partial_\tau \cdot n = \text{tr}(P_\tau \nabla n) \quad \text{on } \Gamma(t).$$

With this convention, a sphere of radius R has curvature $\kappa = (d-1)/R > 0$. The extension of the curvature off the interface is given by

$$(3.12b) \quad \kappa = \Delta d_{\Gamma} \quad \text{for } (x, t) \in \mathcal{N}_{\rho}.$$

It follows from (3.9) that

$$(3.13) \quad n \in C^{\infty}(\mathcal{N}_{\rho}; \mathbb{R}^d), \quad \kappa \in C^{\infty}(\mathcal{N}_{\rho}).$$

3.1.3. Normal velocity and jump notation. The normal velocity of $\Gamma_j(t)$ is denoted by

$$(3.14) \quad V_j(\alpha, t) := \partial_t X_j(\alpha, t) \cdot n(X_j(\alpha, t), t),$$

so that $V_j > 0$ corresponds to motion into the exterior phase. The signed-distance regularity (3.9) allows us to differentiate $d_{\Gamma}(X_j(\alpha, t), t) = 0$ to obtain

$$(3.15) \quad \partial_t d_{\Gamma}(X_j(\alpha, t), t) = -V_j(\alpha, t).$$

Finally, if a quantity g has traces on both sides of the interface, we write

$$(3.16) \quad g^{\pm}(X, t) := \lim_{h \downarrow 0} g(X \pm hn(X, t), t), \quad [g] := g^{+} - g^{-}.$$

3.2. The Mullins–Sekerka limit system. Under the regularity and separation hypotheses above, the signed-distance coordinates furnished by Lemma 3.3 provide the geometric framework for a formal passage to the limit $\varepsilon \rightarrow 0$ in the Cahn–Hilliard system (2.2). The Modica–Mortola theorem [30, 29] identifies the limiting diffuse energy with $\sigma|\Gamma(t)|$, where $|\Gamma(t)|$ is the total interfacial measure (3.3) and σ is the interfacial energy (2.11) associated with the equilibrium profile q .

Using matched asymptotic expansions, Pego [33] showed that the chemical potential admits the formal leading-order limit in each bulk phase,

$$(3.17) \quad \lim_{\varepsilon \rightarrow 0} \mu_{\varepsilon}(x, t) = \widehat{\mu}_0^{\pm}(x, t), \quad \text{locally in } D^{\pm}(t),$$

where $\widehat{\mu}_0$ is governed by the Mullins–Sekerka system (3.20) below. In the bulk, the leading phase field takes the constant values $\phi_0^{\pm} = \pm 1$, so that its time derivative vanishes and the leading-order mass-conservation equation reduces to $\Delta \widehat{\mu}_0^{\pm} = 0$. Near the interface, the signed-distance coordinate in (3.10) is rescaled by setting $z = d_{\Gamma}(x, t)/\varepsilon$. In these inner coordinates, the leading-order phase field is the equilibrium profile $q(z)$ in (2.10), which connects the two bulk values. Matching the inner and outer expansions yields continuity of $\widehat{\mu}_0$ across the interface, with the trace given by the so-called Gibbs–Thomson condition. The inner mass balance gives the normal velocity in terms of $[\partial_n \widehat{\mu}_0]$, so that the jump in the normal derivative drives the interface motion.

At leading order, conservation of the diffuse phase mass reduces to conservation of the total volume of $D^{-}(t)$. Enforcing this global constraint introduces a time-dependent scalar Lagrange multiplier, which we denote by $\mu^f(t)$, into the Gibbs–Thomson condition:

$$(3.18) \quad \widehat{\mu}_0^{-} = \widehat{\mu}_0^{+} = -\frac{\sigma}{2}\kappa - \mu^f(t) \quad \text{on } \Gamma(t).$$

It is convenient to absorb this scalar into the limiting chemical potential by defining

$$(3.19) \quad \mu_0^{\pm} := \widehat{\mu}_0^{\pm} + \mu^f(t),$$

which yields the standard Gibbs–Thomson condition (3.20b). Moreover, (2.5) becomes the far-field condition (3.20d). The resulting sharp-interface system is summarized as follows.

THEOREM 3.4 (Formal sharp-interface limit [33]). *Let $(\phi_\varepsilon, \mu_\varepsilon)$ be a family of smooth solutions of (2.2). Assume the mobility scaling (3.1), matched inner and outer expansions about $\Gamma(t)$, and that the smooth interfaces satisfy Assumption 3.2 throughout $[0, T]$. Then, formally at leading order as $\varepsilon \rightarrow 0$, $\phi_\varepsilon \rightarrow \phi_0$, where ϕ_0 is given by (3.4), and $\mu_\varepsilon \rightarrow \mu_0^\pm - \mu^f(t)$ locally in $D^\pm(t)$, where*

$$(3.20a) \quad \Delta \mu_0^\pm = 0 \quad \text{in } D^\pm(t),$$

$$(3.20b) \quad \mu_0^- = \mu_0^+ = -\frac{\sigma}{2}\kappa \quad \text{on } \Gamma(t),$$

$$(3.20c) \quad V_j = -\frac{m_0}{2}[\partial_n \mu_0] \quad \text{on } \Gamma_j(t), \quad j = 1, \dots, N,$$

$$(3.20d) \quad \mu_0^+(x, t) \rightarrow \mu^f(t) \quad \text{as } |x| \rightarrow \infty.$$

Here σ is defined in (2.11), and $\mu^f \in C^\infty([0, T])$ is the far-field Lagrange multiplier enforcing global mass conservation.

For well-prepared initial data, Alikakos–Bates–Chen [2] rigorously justify Theorem 3.4 on any fixed time interval over which the Mullins–Sekerka evolution (3.20) admits a smooth classical solution. The existence and uniqueness of such solutions are given by the following result due to Chen–Hong–Yi [9].³

THEOREM 3.5 (Strong Mullins–Sekerka solutions [9]). *Let $r \in (0, 1)$, and suppose the initial interface $\Gamma(0)$ is a compact, embedded hypersurface satisfying*

$$(3.21) \quad \Gamma(0) \in C^{3+r}.$$

Then there exists $T > 0$ such that (3.20) admits a unique classical solution on $[0, T]$, with the space–time interface belonging to the class

$$(3.22a) \quad \bigcup_{0 \leq t \leq T} (\Gamma(t) \times \{t\}) \in C^{3+r, (3+r)/3}.$$

Moreover, for every $s \in (0, T)$,

$$(3.22b) \quad \bigcup_{s \leq t \leq T} (\Gamma(t) \times \{t\}) \in C^\infty.$$

Here C^r and $C^{r, r/3}$ denote the spatial and anisotropic space–time Hölder spaces, respectively (see [9] for precise definitions).

Remark 3.6. The instantaneous smoothing property (3.22b) ensures that, for every $s \in (0, T)$, the evolving interfaces admit parametrizations satisfying (3.2c) on $[s, T]$, thereby justifying the smoothness assumed in the matched-asymptotic expansion.

Remark 3.7. The conservation and gradient-flow structure of the diffuse system (cf. Lemma 2.2) pass formally to the sharp-interface limit as conservation of the enclosed volume and dissipation of the total interfacial area.

3.3. Boundary integral formulation. At each fixed time, the bulk harmonic problem in (3.20) can be eliminated in favor of an integral equation posed only on the moving interface using a single-layer potential representation.

³This is a slight restatement of Theorem 1.1 in [9], which is formulated on a bounded domain with homogeneous Neumann boundary conditions. We use the corresponding whole-space formulation.

PROPOSITION 3.8 (Boundary-integral formulation [7, 42]). *Let $\Gamma(t)$ be a family of smooth disjoint closed interfaces. We denote the fundamental solution of $-\Delta$ by*

$$(3.23) \quad \mathcal{G}(x) = \begin{cases} -\frac{1}{2\pi} \log |x|, & d = 2, \\ \frac{1}{4\pi|x|}, & d = 3, \end{cases} \quad x \neq 0.$$

For a smooth density λ on $\Gamma(t)$, define the single-layer potential

$$(3.24) \quad \mathcal{S}_{\Gamma(t)}[\lambda](x) := \int_{\Gamma(t)} \mathcal{G}(x-y) \lambda(y) \, dS_y.$$

Let V denote the normal velocity on $\Gamma(t)$, defined by

$$V(X_j(\alpha, t), t) := V_j(\alpha, t).$$

Then the Mullins–Sekerka system (3.20) is equivalent to the coupled boundary-integral system

$$(3.25a) \quad \mu^f(t) + \frac{2}{m_0} \mathcal{S}_{\Gamma(t)}[V](x) = -\frac{\sigma}{2} \kappa(x, t), \quad x \in \Gamma(t),$$

$$(3.25b) \quad \int_{\Gamma(t)} V(y, t) \, dS_y = 0.$$

For a given interface configuration, (3.25) determines V and $\mu^f(t)$ simultaneously. The chemical potential is recovered from

$$(3.26) \quad \mu_0(x, t) = \mu^f(t) + \frac{2}{m_0} \mathcal{S}_{\Gamma(t)}[V](x), \quad x \in \mathbb{R}^d \setminus \Gamma(t),$$

and the interface parametrizations evolve according to

$$(3.27) \quad \partial_t X_j(\alpha, t) = V_j(\alpha, t) n(X_j(\alpha, t), t) + V_j^\tau(\alpha, t), \quad V_j^\tau(\alpha, t) \in T_{X_j(\alpha, t)} \Gamma_j(t),$$

for $j = 1, \dots, N$. The tangential velocities V_j^τ may be chosen freely.

Proof. We prove the equivalence at each fixed time. For the forward implication, suppose that (Γ, μ_0) is a smooth solution of (3.20). Since μ_0 is continuous across $\Gamma(t)$ and harmonic in the two bulk phases, its distributional Laplacian is

$$(3.28) \quad \Delta \mu_0 = [\partial_n \mu_0] \delta_{\Gamma(t)}, \quad \langle \delta_{\Gamma(t)}, \psi \rangle := \int_{\Gamma(t)} \psi \, dS.$$

The velocity condition (3.20c) therefore gives

$$(3.29) \quad -\Delta \mu_0 = \frac{2}{m_0} V \delta_{\Gamma(t)}.$$

The single-layer potential (3.24) is harmonic away from the interface and continuous across it, while its normal derivative satisfies the standard jump relation

$$(3.30) \quad [\partial_n \mathcal{S}_{\Gamma(t)}[\lambda]] = -\lambda.$$

Consequently, the solution of (3.29) satisfying the far-field condition (3.20d) is given by (3.26). Taking its trace on $\Gamma(t)$ and imposing the Gibbs–Thomson condition

(3.20b) yields (3.25a). Global mass conservation requires the total volume of $D^-(t)$ to remain constant. The transport identity therefore gives

$$0 = \frac{d}{dt}|D^-(t)| = \int_{\Gamma(t)} V(y, t) dS_y,$$

which is precisely (3.25b). In the coupled system (3.25), the scalar $\mu^f(t)$ is the Lagrange multiplier selected simultaneously with V so that this constraint is satisfied. Finally, decomposing $\partial_t X_j$ into its normal and tangential components gives (3.27).

For the converse implication, suppose that V and $\mu^f(t)$ satisfy (3.25) and that the interface parametrizations evolve according to (3.27). Define μ_0 by (3.26). The single-layer potential is harmonic away from $\Gamma(t)$, so (3.20a) holds, and its continuity across the interface, together with (3.25a), gives the Gibbs–Thomson condition (3.20b). Moreover, the jump relation (3.30) gives the velocity condition (3.20c). In three dimensions, the constraint (3.25b) removes the leading monopole term in the expansion of (3.23), whereas in two dimensions it removes the leading logarithmic term. Consequently, $\mathcal{S}_{\Gamma(t)}[V](x) = \mathcal{O}(|x|^{-2})$ when $d = 3$ and $\mathcal{S}_{\Gamma(t)}[V](x) = \mathcal{O}(|x|^{-1})$ when $d = 2$, as $|x| \rightarrow \infty$. Hence (3.20d) also holds, completing the equivalence. $\square$

Remark 3.9. The equivalence in Proposition 3.8 is understood at the level of smooth solutions. For the forward implication, the required regularity is provided by Theorem 3.5. For the converse implication, Chen [7] has established local-in-time existence for the direct boundary-integral evolution of a single interface in two dimensions; to the best of our knowledge, uniqueness and regularity of such solutions, along with the three-dimensional case, remain open.

Remark 3.10. In two dimensions, the tangential velocity in (3.27) may be chosen to preserve equal-arclength parametrizations in the so-called θ - L formulation [21]. The boundary-integral equations in this formulation can be solved accurately and efficiently by pairing Fourier spectral discretization in space with semi-implicit time integration of the stiff leading-order terms, see [21, 42] and Section 7.

4. Intermittent and isolated topological transitions. To summarize, two PDE systems have been introduced thus far: the Cahn–Hilliard system (2.2) and the Mullins–Sekerka system (3.20). Under Assumption 3.2, Theorem 3.4 identifies the latter as the formal sharp-interface limit of the former. For sufficiently regular initial interfaces, Theorem 3.5 gives a unique local classical solution of this limit system, and Proposition 3.8 gives its equivalent boundary integral formulation, for which efficient and accurate numerical solvers are available. Taken together, these results justify the classical sharp-interface evolution while the transition layers remain separated. Consequently, they do not permit topological events such as coalescence.

The Alikakos–Bates–Chen result [2] is restricted to precisely this classical regime, since it assumes the existence of a uniform tubular neighborhood on which the signed-distance function is smooth. When Assumption 3.2 fails, the medial axis enters the transition layer, the closest-point projection ceases to be unique, and the signed-distance coordinates underlying the matched-asymptotic construction break down.

4.1. Weak continuation through varifold solutions. One way to interpret the continuation of the sharp-interface evolution beyond a topological singularity is to adopt a weaker solution concept. This is the approach taken by Chen [8], who proved that a subsequence of Cahn–Hilliard solutions converges as $\varepsilon \rightarrow 0$ to a global varifold solution of the Mullins–Sekerka system, which remains meaningful without a smooth

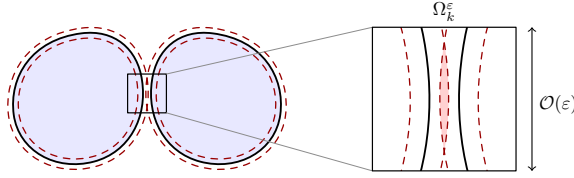


FIG. 3. **Localized failure of the reach condition** (3.6). The solid black curves represent two approaching sharp interfaces, while the dashed red curves delimit their diffuse transition layers. The inset magnifies an $\mathcal{O}(\varepsilon)$ neighborhood Ω_k^ε in which the layers overlap and the closest-point projection becomes nonunique along the medial axis.

embedded interface. Such a weak (measure-valued) continuation does not, however, provide a unique smooth interface from which to restart the classical evolution.

4.2. Strong continuation through interface surgery. An alternative program, outlined by Lowengrub and Truskinovsky [27], seeks to resolve topological transitions through diffuse mixing dynamics localized in both space and time. While the transition layers remain well separated, their internal structure may be neglected and the sharp-interface system governs the evolution. When two layers overlap, however, the diffuse mixing dynamics must be resolved. These dynamics provide a smooth continuation through the topological singularity, after which a new separated sharp interface emerges. In this picture, the regularization and its associated energy dissipation are localized in space and time near the topological event.

The strong continuation of the solution is realized here through a formal *interface surgery* algorithm, which will be described below. At a topological event time T , its purpose is to take the incoming interface family immediately before the event $t = T^-$ and return the outgoing interface family immediately after the event $t = T^+$:

$$(4.1) \quad \mathbf{\Gamma}(T^\pm) = \bigcup_{j=1}^{N(T^\pm)} \Gamma_j(T^\pm), \quad \Gamma_j(T^\pm) = X_j(\mathbb{S}^{d-1}, T^\pm).$$

Note that the component counts $N(T^-)$ and $N(T^+)$ need not agree.

4.3. Topological events. Let $r \in (0, 1)$, and let $\mathbf{\Gamma}(0)$ be a compact, embedded C^{3+r} hypersurface satisfying the reach condition (3.6). By Theorem 3.5, the initial interface generates a unique local classical solution $\mathbf{\Gamma}(t)$ of the Mullins–Sekerka system. On any interval where Assumption 3.2 holds, Theorem 3.4 justifies using this solution as the leading-order approximation to the diffuse-interface evolution. Finally, Proposition 3.8 supplies the equivalent boundary integral formulation used to compute this classical solution. Let us consider the evolution on the time interval

$$0 \leq t \leq t_{\max}, \quad t_{\max} > 0.$$

DEFINITION 4.1 (Topological event times). *The first topological event time is*

$$(4.2a) \quad T_0 := \inf \{t \in (0, t_{\max}] : \text{reach}(\mathbf{\Gamma}(t)) \leq C_{\text{layer}}\varepsilon\}.$$

If $T_0 > t_{\max}$, no topological transition occurs. Otherwise, define the incoming interface

$$\mathbf{\Gamma}(T_0^-) := \lim_{t \uparrow T_0} \mathbf{\Gamma}(t).$$

Once the surgery constructs the outgoing interface $\Gamma(T_k^+)$ and the classical evolution is restarted from these data, the next event time is defined inductively by

$$(4.2b) \quad T_{k+1} := \inf \{t \in (T_k, t_{\max}] : \text{reach}(\Gamma(t)) \leq C_{\text{layer}}\varepsilon\}.$$

If this set is nonempty, set

$$\Gamma(T_{k+1}^-) := \lim_{t \uparrow T_{k+1}} \Gamma(t).$$

Thus, on every open interval (T_k^+, T_{k+1}^-) , the interface evolves by the classical Mullins–Sekerka system. The sharp-interface evolution is suspended only at the times T_k when the separated-layer condition fails. At such a time, the loss of separation is localized by identifying points in the effective transition layer with more than one nearest point on the interface.

DEFINITION 4.2 (Reach-defect set). *For a closed interface Γ , define the nearest-point set and medial axis by*

$$(4.3a) \quad \Pi_\Gamma(x) := \{y \in \Gamma : |x - y| = \text{dist}(x, \Gamma)\},$$

$$(4.3b) \quad \text{Med}(\Gamma) := \{x \in \mathbb{R}^d : \#\Pi_\Gamma(x) > 1\}.$$

At the event time T_k , define the reach-defect set by

$$(4.3c) \quad \mathcal{Z}_k^\varepsilon := \{x \in \text{Med}(\Gamma(T_k^-)) : \text{dist}(x, \Gamma(T_k^-)) \leq C_{\text{layer}}\varepsilon\}.$$

It contains the points at which the closest-point projection ceases to be single-valued within the effective transition layer.

We consider topological events that are *intermittent* in time and *isolated* in space. Intermittency means that only finitely many events occur and that successive events remain separated on the macroscopic time scale. Spatial isolation means that each reach-defect set is contained in a bounded region whose diameter is $\mathcal{O}(\varepsilon)$. This models a point-like coalescence or pinch-off in which the defect-region length scale is of size ε , i.e., a *microscopic defect*. The corresponding d -dimensional volume is $\mathcal{O}(\varepsilon^d)$. Extended near-contact regions and simultaneous nonlocal topology changes are excluded. By Theorem 3.5, each classical branch becomes smooth instantaneously. Since successive event times remain separated, we may therefore take every incoming interface to be a finite union of pairwise disjoint, C^∞ -smooth, embedded hypersurfaces. These requirements are collected in the following assumption.

ASSUMPTION 4.3 (Intermittent and isolated transitions). *There exist an integer $K \in \mathbb{N}_0$ and constants $\tau_* > 0$, $C_* > 0$, and $c_{\text{buf}} > 0$, independent of ε , such that*

$$(4.4a) \quad 0 < T_0 < T_1 < \cdots < T_K < t_{\max}, \quad T_{k+1} - T_k \geq \tau_*, \quad 0 \leq k < K.$$

For each $0 \leq k \leq K$, the number of interfaces is finite, $N(T_k^-) < \infty$, and

$$(4.4b) \quad X_j(\cdot, T_k^-) : \mathbb{S}^{d-1} \longrightarrow \mathbb{R}^d \text{ is a } C^\infty \text{ embedding, } \quad 1 \leq j \leq N(T_k^-),$$

with pairwise disjoint components:

$$(4.4c) \quad \Gamma_j(T_k^-) \cap \Gamma_\ell(T_k^-) = \emptyset, \quad 1 \leq j, \ell \leq N(T_k^-), \quad j \neq \ell.$$

Moreover, there exists a bounded open defect region Ω_k^ε such that

$$(4.4d) \quad \overline{\mathcal{Z}_k^\varepsilon} \subset \Omega_k^\varepsilon, \quad \text{diam}(\Omega_k^\varepsilon) \leq C_*\varepsilon, \quad \text{dist}(\overline{\mathcal{Z}_k^\varepsilon}, \partial\Omega_k^\varepsilon) \geq c_{\text{buf}}\varepsilon.$$

Remark 4.4 (Multiple defect regions). Assumption 4.3 extends directly to finitely many simultaneous, spatially separated defects. If

$$\mathcal{Z}_k^\varepsilon = \bigcup_{\ell=1}^{L_k} \mathcal{Z}_{k,\ell}^\varepsilon, \quad \Omega_k^\varepsilon = \bigcup_{\ell=1}^{L_k} \Omega_{k,\ell}^\varepsilon, \quad L_k < \infty,$$

where the $\mathcal{Z}_{k,\ell}^\varepsilon$ are the connected components of $\mathcal{Z}_k^\varepsilon$ and the open sets $\Omega_{k,\ell}^\varepsilon$ have pairwise disjoint closures, we require each pair $(\mathcal{Z}_{k,\ell}^\varepsilon, \Omega_{k,\ell}^\varepsilon)$ to satisfy (4.4d). If L_k is bounded independently of ε , then $|\Omega_k^\varepsilon| = \mathcal{O}(\varepsilon^d)$.

5. Localized pseudo-time relaxation. At each topological event, we suspend the sharp-interface evolution and resolve the overlapping transition layers in an $\mathcal{O}(\varepsilon)$ neighborhood of the reach-defect set. This neighborhood is rescaled to a fixed patch, where a mass-conserving localized Cahn–Hilliard flow is evolved in fast pseudo-time while the exterior interface remains fixed.

In the following, we suppress the indices ε and k from the reach-defect set and defect region, writing $\mathcal{Z}$ and Ω for $\mathcal{Z}_k^\varepsilon$ and Ω_k^ε , respectively. We take Ω to be a canonical bounded open domain containing the entire reach-defect set. More precisely, we choose a center $x_c \in \mathbb{R}^d$ and a constant $L > 0$, independent of ε , and set

$$(5.1a) \quad \Omega = x_c + \varepsilon(-L, L)^d.$$

Thus, Ω is a square in two dimensions and a cube in three dimensions. In accordance with (4.4d), it is chosen so that

$$(5.1b) \quad \overline{\mathcal{Z}} \subset \Omega, \quad \text{dist}(\overline{\mathcal{Z}}, \partial\Omega) \geq c_{\text{buf}}\varepsilon, \quad \text{diam}(\Omega) \leq C_*\varepsilon, \quad |\Omega| \leq C_*\varepsilon^d.$$

DEFINITION 5.1 (Defect-region cutoff). *A defect-region cutoff is a smooth compactly supported function $\eta \in C_c^\infty(\Omega)$ satisfying*

$$(5.2) \quad 0 \leq \eta \leq 1 \quad \text{in } \Omega, \quad \eta \equiv 1 \quad \text{near } \overline{\mathcal{Z}}, \quad \|\partial_x^r \eta\|_{L^\infty(\Omega)} \leq C_r \varepsilon^{-r}, \quad r \geq 0,$$

where the constants C_r are independent of ε .

5.1. Patch coordinates. The physical diameter of Ω is $\mathcal{O}(\varepsilon)$. We therefore re-center the patch at x_c and measure distances in units of ε . This maps the microscopic physical patch to a fixed domain of order-one size.

DEFINITION 5.2 (Patch coordinates). *The patch coordinate and rescaled defect domain are*

$$(5.3) \quad \xi := \frac{x - x_c}{\varepsilon}, \quad x = x_c + \varepsilon\xi, \quad \text{and} \quad Q := \frac{\Omega - x_c}{\varepsilon} = (-L, L)^d.$$

The rescaled reach-defect set and cutoff are

$$(5.4) \quad \mathcal{Y} := \frac{\mathcal{Z} - x_c}{\varepsilon} \quad \text{and} \quad \chi(\xi) := \eta(x_c + \varepsilon\xi), \quad \xi \in Q.$$

The chain rule and (5.2) imply that

$$\partial_\xi^r \chi(\xi) = \varepsilon^r \partial_x^r \eta(x_c + \varepsilon\xi), \quad \|\partial_\xi^r \chi\|_{L^\infty(Q)} \leq C_r, \quad r \geq 0.$$

Moreover, (5.1b) gives

$$\text{dist}(\overline{\mathcal{Y}}, \partial Q) \geq c_{\text{buf}}.$$

Finally, since $\chi \in C_c^\infty(Q)$, its support is a compact subset of Q and therefore lies a positive distance from ∂Q . Set

$$(5.5a) \quad \delta_\chi := \frac{1}{2} \text{dist}(\text{supp } \chi, \partial Q) > 0,$$

and define the corresponding boundary collar by

$$(5.5b) \quad \mathcal{C}_\chi := \{\xi \in Q : \text{dist}(\xi, \partial Q) < \delta_\chi\} \subset Q \setminus \text{supp } \chi \subset \{\chi = 0\}.$$

5.2. Pseudo-time scaling. The spatial rescaling determines the corresponding fast temporal scale. At the topological event time T , we introduce the inner phase field and chemical potential

$$(5.6a) \quad \Phi(\xi, s) := \phi_\varepsilon(x_c + \varepsilon\xi, T + t_{\text{mix}}s),$$

$$(5.6b) \quad U(\xi, s) := \frac{\varepsilon}{\widehat{\sigma}} \mu_\varepsilon(x_c + \varepsilon\xi, T + t_{\text{mix}}s).$$

Here, s is the dimensionless pseudo-time, while t_{mix} is its physical timescale. By the chain rule, (2.2a) becomes

$$\partial_s \Phi = \frac{m_\varepsilon \widehat{\sigma} t_{\text{mix}}}{\varepsilon^3} \Delta_\xi U.$$

The two sides balance at order one when

$$(5.7) \quad t_{\text{mix}} = \frac{\varepsilon^3}{m_\varepsilon \widehat{\sigma}}, \quad s = \frac{m_\varepsilon \widehat{\sigma}}{\varepsilon^3} (t - T).$$

In these variables, the inner Cahn–Hilliard system becomes

$$(5.8a) \quad \partial_s \Phi = \Delta_\xi U,$$

$$(5.8b) \quad U = -\Delta_\xi \Phi + W'(\Phi).$$

Thus, an $\mathcal{O}(1)$ pseudo-time interval corresponds to an $\mathcal{O}(\varepsilon^3)$ physical-time interval. An order-one macroscale velocity displaces the interface by only $\mathcal{O}(\varepsilon^3)$ during this interval, which is negligible relative to the $\mathcal{O}(\varepsilon)$ defect-region length scale. The macroscale sharp-interface evolution may therefore be frozen at $t = T$ during the localized pseudo-time evolution.

The spatial rescaling also transforms the free energy. The contribution from the physical defect region is

$$(5.9) \quad E_{\varepsilon, \Omega}[\phi_\varepsilon] := \widehat{\sigma} \int_\Omega \left(\frac{\varepsilon}{2} |\nabla_x \phi_\varepsilon|^2 + \frac{1}{\varepsilon} W(\phi_\varepsilon) \right) dx = \widehat{\sigma} \varepsilon^{d-1} \mathcal{F}[\Phi],$$

where $\mathcal{F}[\Phi]$ is the rescaled free energy associated with (5.8),

$$(5.10) \quad \mathcal{F}[\Phi] := \int_Q \left(\frac{1}{2} |\nabla_\xi \Phi|^2 + W(\Phi) \right) d\xi.$$

5.3. Localized dissipation form. The system (5.8) has the same gradient-flow structure as (2.2). More precisely, it is the $H^{-1}(Q)$ -gradient flow of the rescaled free energy (5.10). Here, the $H^{-1}(Q)$ inner product is defined as in (2.18), with the

whole-space operator $(-\Delta)^{-1}$ replaced by the inverse Neumann Laplacian on zero-mean functions in Q . The dissipation law for (5.8) is

$$(5.11) \quad \frac{d}{ds} \mathcal{F}[\Phi] = - \int_Q |\nabla_\xi U|^2 d\xi.$$

Thus, the classical local Cahn–Hilliard evolution (5.8) dissipates energy wherever $\nabla_\xi U \neq 0$ in Q . The interface outside Q remains fixed; if dissipation extends to ∂Q , the zero level set will move and therefore fail to join smoothly to the unchanged exterior interface. To restrict the dissipative effects to the defect region, we employ the cutoff (5.4) and introduce the following symmetric bilinear form.

DEFINITION 5.3 (Localized dissipation form). *For smooth functions $f, g : Q \rightarrow \mathbb{R}$, define*

$$(5.12a) \quad \mathbf{a}_\chi(f, g) := \int_Q \nabla_\xi(\chi f) \cdot \nabla_\xi(\chi g) d\xi.$$

The associated localized dissipation seminorm is

$$(5.12b) \quad |f|_\chi := \mathbf{a}_\chi(f, f)^{1/2} = \|\nabla_\xi(\chi f)\|_{L^2(Q)}.$$

The form (5.12a) is symmetric and positive semidefinite, while (5.12b) defines only a seminorm. The resulting degeneracy will freeze the evolution where $\chi = 0$. By (5.2), $\chi = 1$ on the defect core, recovering the usual Cahn–Hilliard dissipation.

5.4. Mass-conserving localized evolution. Next, we use the localized dissipation form (5.12) to construct a mass-conserving gradient flow whose dissipation is confined to the reach-defect region. Let $\Phi_{\text{in}} := \Phi(\cdot, 0)$. By (2.13),

$$\Phi_{\text{in}}(\xi) = q(D_{\text{in}}(\xi)), \quad D_{\text{in}}(\xi) := \frac{1}{\varepsilon} d_{\Gamma(T-)}(x_c + \varepsilon \xi).$$

It follows that

$$|\Phi_{\text{in}}(\xi) - \Phi_{\text{in}}(\zeta)| \leq \|q'\|_{L^\infty} |\xi - \zeta|,$$

so that $\Phi_{\text{in}} \in W^{1,\infty}(Q) \subset H^1(Q)$. On the rescaled reach-defect set $\mathcal{Y}$, however, the closest-point projection is not unique, so D_{in} and Φ_{in} are generally not continuously differentiable. The cutoff then restricts the set of admissible configurations to

$$(5.13) \quad \mathcal{X} := \{\Psi \in H^1(Q) : \Psi = \Phi_{\text{in}} \text{ a.e. on } \{\chi = 0\}\}.$$

The pseudo-time evolution is associated with the constrained minimization problem

$$(5.14) \quad \min_{\Psi \in \mathcal{X}} \mathcal{F}[\Psi] \quad \text{subject to} \quad \int_Q \Psi d\xi = \int_Q \Phi_{\text{in}} d\xi.$$

The corresponding Lagrangian is

$$(5.15) \quad \mathcal{L}[\Psi, \lambda] := \mathcal{F}[\Psi] + \lambda \left(\int_Q \Psi d\xi - \int_Q \Phi_{\text{in}} d\xi \right),$$

where $\lambda \in \mathbb{R}$ is the Lagrange multiplier for the phase-mass constraint. Since the variational derivative of (5.15) is $U + \lambda$, the localized constrained gradient flow is defined weakly by

$$(5.16) \quad \int_Q \partial_s \Phi \psi d\xi = -\mathbf{a}_\chi(U + \lambda, \psi), \quad \psi \in H^1(Q).$$

Here, $\lambda = \lambda(s)$ is chosen so that the flow remains on the constraint set in (5.14). Differentiating the constraint gives

$$\frac{d}{ds} \int_Q \Phi \, d\xi = 0 \implies \mathfrak{a}_\chi(U + \lambda, 1) = 0.$$

Since the cutoff is nonconstant,

$$\mathfrak{a}_\chi(1, 1) = \int_Q |\nabla_\xi \chi|^2 \, d\xi > 0,$$

so the multiplier is uniquely determined as

$$(5.17) \quad \lambda(s) = -\frac{\mathfrak{a}_\chi(U, 1)}{\mathfrak{a}_\chi(1, 1)}.$$

5.5. Localized pseudo-time system. We now write the localized weak flow (5.16) in strong form. Define the localized chemical potential

$$(5.18) \quad Z := (U + \lambda)\chi.$$

Since $\chi \in C_c^\infty(Q)$, the product $\chi\psi$ belongs to $H_0^1(Q)$. Assuming sufficient regularity, integration by parts in (5.16) gives

$$-\mathfrak{a}_\chi(U + \lambda, \psi) = - \int_Q \nabla_\xi Z \cdot \nabla_\xi (\chi\psi) \, d\xi = \int_Q \chi \Delta_\xi Z \, \psi \, d\xi.$$

Consequently, the localized pseudo-time evolution is

$$(5.19a) \quad \partial_s \Phi = \chi \Delta_\xi Z, \quad \text{in } Q \times (0, s_{\max}),$$

$$(5.19b) \quad Z = (U + \lambda)\chi, \quad \text{in } Q \times (0, s_{\max}),$$

$$(5.19c) \quad U = -\Delta_\xi \Phi + W'(\Phi), \quad \text{in } Q \times (0, s_{\max}),$$

$$(5.19d) \quad \lambda(s) = -\frac{\int_Q \nabla_\xi (\chi U) \cdot \nabla_\xi \chi \, d\xi}{\int_Q |\nabla_\xi \chi|^2 \, d\xi}, \quad 0 < s < s_{\max},$$

$$(5.19e) \quad \Phi(\xi, 0) = \Phi_{\text{in}}(\xi), \quad \text{in } Q,$$

$$(5.19f) \quad \Phi(\xi, s) = \Phi_{\text{in}}(\xi), \quad Z(\xi, s) = 0, \quad \text{in } \mathcal{C}_\chi \times (0, s_{\max}).$$

Here, $s_{\max}$ is the stopping pseudo-time, which will be defined below. The collar identities (5.19f) are not independent boundary conditions. Since $\chi = 0$ on $\mathcal{C}_\chi$, (5.19a) gives $\partial_s \Phi = 0$ there, and the initial condition then gives $\Phi = \Phi_{\text{in}}$; similarly, (5.19b) gives $Z = 0$. Thus the localized evolution itself freezes both fields throughout an open neighborhood of ∂Q . For numerical purposes, it is convenient to replace (5.19f) by the weaker implied Neumann conditions,

$$(5.20) \quad \partial_\nu (\Phi - \Phi_{\text{in}}) = 0, \quad \partial_\nu Z = 0, \quad \text{on } \partial Q \times (0, s_{\max}),$$

where ν is the outward unit normal to ∂Q .

Taking $\psi = U + \lambda$ in (5.16) and using the phase-mass constraint gives the localized energy dissipation law

$$(5.21) \quad \frac{d}{ds} \mathcal{F}[\Phi(s)] = -\mathfrak{a}_\chi(U + \lambda, U + \lambda) = - \int_Q |\nabla_\xi Z|^2 \, d\xi =: -\mathcal{D}_\chi(s).$$

5.6. Termination of the localized relaxation. The localized system (5.19) is evolved only while it produces an appreciable decrease of the free energy. For nontrivial interface surgery, the localized free energy is positive, $\mathcal{F}[\Phi(s)] > 0$. We define the relative energy-dissipation rate by

$$(5.22) \quad \mathcal{R}_\chi(s) := \frac{\mathcal{D}_\chi(s)}{\mathcal{F}[\Phi(s)]} = -\frac{d}{ds} \log \mathcal{F}[\Phi(s)],$$

where $\mathcal{D}_\chi$ is defined in (5.21). Thus $\mathcal{R}_\chi$ measures the instantaneous fractional rate at which the free energy decreases.

DEFINITION 5.4 (Terminal pseudo-time). *Fix a dissipation tolerance $\delta_s > 0$. The localized pseudo-time evolution (5.19) is terminated at the terminal pseudo-time*

$$(5.23) \quad s_{\max} := \inf \{s > 0 : \mathcal{R}_\chi(s) \leq \delta_s\}.$$

Let $\mathcal{F}_{\min}$ denote the mass-constrained infimum of the rescaled free energy,

$$(5.24) \quad \mathcal{F}_{\min} := \inf \left\{ \mathcal{F}[\Psi] : \Psi \in \mathcal{X}, \quad \int_Q \Psi \, d\xi = \int_Q \Phi_{\text{in}} \, d\xi \right\}.$$

LEMMA 5.5 (Bound on the terminal pseudo-time). *Suppose that there are constants $c_F, C_F > 0$, independent of ε , such that*

$$(5.25) \quad 0 < c_F \leq \mathcal{F}_{\min} \leq \mathcal{F}[\Phi_{\text{in}}] \leq C_F.$$

Then the terminal pseudo-time $s_{\max}$ defined by (5.23) is bounded above by

$$(5.26) \quad s_{\max} \leq \frac{1}{\delta_s} \log \left(\frac{C_F}{c_F} \right).$$

Consequently, $s_{\max} = \mathcal{O}(1)$ as $\varepsilon \rightarrow 0$.

Proof. For $0 < s < s_{\max}$, (5.23) gives $\mathcal{R}_\chi(s) > \delta_s$. Hence, by (5.22),

$$\mathcal{F}[\Phi(s)] \leq \mathcal{F}[\Phi_{\text{in}}] e^{-\delta_s s}.$$

Combining this inequality with (5.25) gives (5.26). $\square$

6. Interface surgery and sharp-interface restart. At each topological event, the localized system (5.19) is evolved to the terminal pseudo-time $s = s_{\max}$ defined in (5.23), and the terminal phase field $\Phi(\cdot, s_{\max})$ is used to construct an outgoing interface from which the sharp-interface evolution is restarted. A rigorous justification requires two ingredients: (1) the localized system must be globally well posed and relax toward a sufficiently regular, stable equilibrium, and (2) the zero level set $\{\Phi(\cdot, s_{\max}) = 0\}$ must define a smooth outgoing interface that matches the incoming interface outside the defect region and again satisfies (3.6). Complete proofs of these properties lie beyond the scope of this work; below, we outline the basic analytical framework and verify the geometric reconstruction for an idealized coalescence configuration.

6.1. Well-posedness and asymptotic relaxation. The standard Galerkin and energy-estimate arguments for the Cahn–Hilliard system (2.2) (see, e.g., Miranville [28]) can be adapted to establish well-posedness of the localized system (5.19) in a natural weak-solution class. The principal additional difficulty is the cutoff-induced degeneracy: the leading fourth-order coefficient is χ^2 and vanishes where

$\chi = 0$, so the equation is not uniformly parabolic throughout Q . Nevertheless, χ is bounded away from zero on every compact subset of $\{\chi > 0\}$ and is identically one near the rescaled reach-defect set $\mathcal{Y}$. Standard interior regularity therefore applies in the region where the topological transition is resolved.

The dissipation identity (5.21) makes $\mathcal{F}$ a Lyapunov functional for the localized evolution. For the classical Cahn–Hilliard equation, compactness and Łojasiewicz–Simon arguments yield convergence to a single equilibrium [36]. The same general framework provides the natural route to describing the long-time behavior of (5.19), although the cutoff-induced degeneracy must be accommodated in the corresponding functional setting. Suppose that the localized evolution converges to an isolated, stable constrained minimizer Φ_* . If the stopping criterion places $\Phi(\cdot, s_{\max})$ sufficiently close to Φ_* in a topology strong enough to control level-set geometry, then the smoothness, matching, and reach properties of $\{\Phi_* = 0\}$ transfer to the terminal zero level set $\{\Phi(\cdot, s_{\max}) = 0\}$. Thus, proving that the equilibrium zero-level set satisfies the regularity requirement (3.21) and restores the separated-layer condition in Assumption 3.2 would justify the sharp-interface restart, completing the logical chain underlying the sharp–diffuse continuation methodology.

6.2. Smooth interface surgery for idealized coalescence. We examine these two requirements in an idealized symmetric model of interface coalescence. The symmetry permits an explicit computation of the post-coalescence sharp-interface equilibrium. We show that this interface satisfies the reach condition (3.6) and matches the prescribed exterior interface with C^4 regularity at the attachment points.

6.2.1. Incoming idealized interface. We specialize to $d = 2$ and take

$$(6.1) \quad Q = (-L, L)^2.$$

For this idealized calculation, we suppress the cutoff transition and set $\chi \equiv 1$. The localized evolution then becomes the unweighted inner Cahn–Hilliard system

$$(6.2a) \quad \partial_s \Phi = \Delta_\xi U,$$

$$(6.2b) \quad U = -\Delta_\xi \Phi + W'(\Phi),$$

with homogeneous Neumann conditions

$$(6.2c) \quad \partial_\nu \Phi = 0, \quad \partial_\nu U = 0 \quad \text{on } \partial Q.$$

Here the phase mass is conserved automatically, so no additional multiplier is required.

The incoming interface consists of the two graphs

$$(6.3) \quad \Gamma_\pm^{\text{in}} := \{(\xi_1, \pm h(\xi_1)) : -L \leq \xi_1 \leq L\} \subset \overline{Q},$$

where $h \in C^4([-L, L])$ is positive and

$$(6.4) \quad h(\pm L) = H.$$

Thus the four attachment points are $(\pm L, \pm H)$.

LEMMA 6.1 (Neumann conditions at the attachment points). *Let Γ be a smooth extension of $\Gamma_+^{\text{in}} \cup \Gamma_-^{\text{in}}$ through the four attachment points, and suppose that its signed-distance function d_Γ is smooth there. Define*

$$\Phi_{\text{in}} = q(d_\Gamma), \quad U_{\text{in}} = -\Delta_\xi \Phi_{\text{in}} + W'(\Phi_{\text{in}}).$$

If $(\Phi_{\text{in}}, U_{\text{in}})$ satisfies (6.2c), then the incoming branches meet the vertical sides of Q orthogonally and

$$(6.5) \quad h'(\pm L) = 0, \quad h'''(\pm L) = 0.$$

Proof. At an attachment point,

$$\partial_\nu \Phi_{\text{in}} = q'(0) \nabla_\xi d_\Gamma \cdot \nu = q'(0) n \cdot \nu.$$

Since $q'(0) \neq 0$, the first condition in (6.2c) gives $n \cdot \nu = 0$. Thus the interface meets the vertical side orthogonally, so that $h'(\pm L) = 0$. Wherever d_Γ is smooth, $|\nabla_\xi d_\Gamma| = 1$, and hence

$$\begin{aligned} U_{\text{in}} &= -q''(d_\Gamma) - q'(d_\Gamma) \Delta_\xi d_\Gamma + W'(q(d_\Gamma)) \\ &= -q'(d_\Gamma) \Delta_\xi d_\Gamma, \end{aligned}$$

where the profile equation (2.9) was used in the second equality. At an attachment point, $\partial_\nu d_\Gamma = n \cdot \nu = 0$, so

$$\partial_\nu U_{\text{in}} = -q'(0) \partial_\nu \Delta_\xi d_\Gamma.$$

The second condition in (6.2c) therefore gives $\partial_\nu \Delta_\xi d_\Gamma = 0$. Since ν is tangent to the interface at orthogonal contact, this is the vanishing of the tangential derivative of curvature. For a graph satisfying $h' = 0$, that derivative is h''' . $\square$

We impose the remaining geometric conditions as follows.

ASSUMPTION 6.2 (Idealized coalescence). *Let h satisfy the attachment conditions (6.4) and (6.5), and suppose that*

$$(6.6) \quad 0 < h(0) < C_{\text{layer}} < H < L - C_{\text{layer}}.$$

The function h is assumed to be even and to satisfy

$$(6.7) \quad h'(0) = 0, \quad h''(0) > 0,$$

together with the attachment conditions

$$(6.8) \quad h''(\pm L) = -\frac{1}{H}, \quad h^{(4)}(\pm L) = -\frac{3}{H^3}.$$

Finally, we impose the area constraint

$$(6.9) \quad \int_0^L h(\xi_1) d\xi_1 = \frac{\pi}{4} H^2.$$

The two incoming branches therefore have a nondegenerate closest approach at $\xi_1 = 0$ and meet the vertical sides of Q orthogonally at $(\pm L, \pm H)$. Since their separation at the center is $2h(0) < 2C_{\text{layer}}$, their effective transition layers overlap there. The constraint (6.9) equates the area between the incoming graphs with the total area enclosed by the two outgoing semicircles constructed below.

EXAMPLE 6.3 (Explicit polynomial height function). *For prescribed values of L , H , and $h(0)$ satisfying (6.6), we construct an explicit admissible height function. Introduce the dimensionless aspect ratio and central height,*

$$(6.10a) \quad \Lambda := \frac{L}{H}, \quad \alpha := \frac{h(0)}{H}.$$

Set $r := 1 - \xi_1^2/L^2$ and define the height function by

$$(6.10b) \quad h(\xi_1) = H\mathcal{P}(r),$$

where

$$(6.10c) \quad \mathcal{P}(r) = 1 - \frac{\Lambda^2}{8}r^2 - \frac{\Lambda^2}{16}r^3 - \frac{\Lambda^4 + 5\Lambda^2}{128}r^4 + Ar^5 + Br^5(1-r).$$

The terms through order r^4 enforce the attachment conditions. The coefficient A enforces the prescribed central height,

$$(6.10d) \quad A = \alpha - 1 + \frac{\Lambda^4 + 29\Lambda^2}{128},$$

and the coefficient B is determined by the area constraint (6.9),

$$(6.10e) \quad B = \frac{9009}{256} \left(\frac{\pi}{4\Lambda} - 1 + \frac{\Lambda^2}{9} + \frac{\Lambda^4}{315} - \frac{256}{693}A \right).$$

The resulting function h is an even polynomial in ξ_1 of degree at most twelve. It satisfies the attachment conditions (6.4), (6.5), and (6.8), the area constraint (6.9), and the central conditions $h(0) = H\alpha$ and $h'(0) = 0$. The sufficient conditions

$$(6.11) \quad 1 < \Lambda < \frac{5}{2}, \quad \frac{\Lambda^4 + 19\Lambda^2}{32} - 5A + B > 0$$

ensure that h is positive on $[-L, L]$ and that $h''(0) > 0$. Consequently, under (6.11), the profile satisfies all the requirements of Assumption 6.2. The algebraic verification is given in Appendix A.

In particular, the parameter choice

$$(6.12) \quad C_{\text{layer}} = 2, \quad L = 10, \quad H = 7.5, \quad h(0) = 1.5$$

satisfies the scale condition (6.6) and the admissibility conditions (6.11). The corresponding incoming and outgoing interfaces are shown in Figure 4.

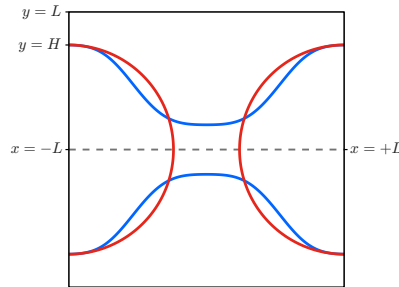


FIG. 4. **Idealized coalescence.** The blue curves are the incoming interface branches $\xi_2 = \pm h(\xi_1)$ generated by the polynomial profile (6.10) with the parameter choice (6.12). The red curves are the reconstructed outgoing equilibrium semicircles (6.13).

6.2.2. Outgoing equilibrium interface. Each component of the outgoing equilibrium interface $\Gamma_{\pm}^{\text{out}}$ has constant curvature and is therefore circular. Symmetry about the horizontal axis, attachment at $(\pm L, \pm H)$, and orthogonal contact with the vertical sides of Q place the centers at $(\pm L, 0)$ and fix the radius to be H . Thus the outgoing interface consists of the two inward-facing semicircles

$$(6.13) \quad \Gamma_{\pm}^{\text{out}} := \{(\xi_1, \xi_2) \in \overline{Q} : (\xi_1 \mp L)^2 + \xi_2^2 = H^2\},$$

shown in red in Figure 4. Each semicircle has radius H , and the distance between the two components is $2(L - H)$. Consequently,

$$(6.14) \quad \text{reach}(\Gamma_{-}^{\text{out}} \cup \Gamma_{+}^{\text{out}}) = \min\{H, L - H\} > C_{\text{layer}},$$

where the final inequality follows from (6.6). Under the physical rescaling $x = x_c + \varepsilon \xi$, the reach is multiplied by ε , producing the required bound (3.6). The matching conditions (6.4), (6.5), and (6.8) then ensure that the outgoing arcs join the prescribed exterior interface with C^4 regularity.

The reconstructed global outgoing interface is therefore a compact, embedded C^4 hypersurface satisfying the separated-layer condition. In particular, it belongs to C^{3+r} for every $r \in (0, 1)$ and hence provides admissible initial data for Theorem 3.5; the sharp-interface evolution can then be restarted from this interface.

7. Some details of the numerical implementation. We now describe our simple two-dimensional implementation of the event-driven sharp-diffuse continuation procedure.⁴ Between topological events, the Mullins-Sekerka evolution is computed using the boundary-integral method of Zhu-Chen-Hou [42]; during each localized relaxation, the Cahn-Hilliard system is discretized using an SAV scheme adapted from Shen-Xu-Yang [37].

7.1. Overview of the hybrid algorithm. The computation alternates between sharp-interface evolution on the full domain and localized diffuse relaxation near detected topological events. The complete event-driven procedure is summarized in Algorithm 7.1.

7.2. Boundary-integral discretization. For the sharp-interface evolution, we use the θ - L boundary-integral method developed in [21, 42]. Each interface component is represented by

$$(7.1a) \quad \partial_{\alpha} X_j(\alpha, t) = \frac{L_j(t)}{2\pi} \begin{pmatrix} \cos \theta_j(\alpha, t) \\ \sin \theta_j(\alpha, t) \end{pmatrix},$$

$$(7.1b) \quad \theta_j(\alpha, t) = \alpha + \psi_j(\alpha, t),$$

where $\alpha \in [0, 2\pi]$, each component is oriented counterclockwise, $L_j(t)$ is its length, and ψ_j is 2π -periodic. In particular, since $|\partial_{\alpha} X_j(\alpha, t)| = L_j(t)/2\pi$, it follows that α is an equal-arclength parameter. The variables θ_j and L_j determine the shape of $\Gamma_j(t)$ but not its position. We therefore also evolve the reference point $a_j(t) := X_j(0, t)$ and reconstruct the interface from

$$(7.1c) \quad X_j(\alpha, t) = a_j(t) + \frac{L_j(t)}{2\pi} \int_0^{\alpha} \begin{pmatrix} \cos \theta_j(\beta, t) \\ \sin \theta_j(\beta, t) \end{pmatrix} d\beta.$$

⁴The code was written in MATLAB, and all numerical experiments were performed on a MacBook Pro equipped with an Apple M1 Pro processor and 32 GB of memory.

Algorithm 7.1 Event-driven hybrid sharp–diffuse algorithm

- Step 1.** *Sharp-interface evolution.* Advance the interface using the boundary-integral formulation (3.25) and (3.27) while the discrete reach criterion remains satisfied.
- Step 2.** *Event detection and localization.* When the reach-defect detector described in Algorithm 7.2 flags an impending topological event, record the incoming interface $\Gamma(T^-)$ and construct the defect patch, cutoff, and rescaled coordinates according to (5.1), Definition 5.1, and Definition 5.2.
- Step 3.** *Localized relaxation.* Freeze the sharp-interface evolution and initialize Φ_{in} from the signed-distance profile (2.13) in the patch coordinates (5.3). Discretize the localized system (5.19) using a finite-volume SAV scheme and evolve until the stopping criterion (5.23) is reached.
- Step 4.** *Interface reconstruction.* Extract the terminal zero level set $\{\Phi(\cdot, s_{\text{max}}) = 0\}$ using marching squares, join it to the unchanged exterior interface, and reparametrize the resulting closed curves for the boundary-integral solver.
- Step 5.** *Sharp-interface restart.* Advance the physical time according to (5.7), verify the outgoing separated-layer condition, and restart the sharp-interface evolution from $\Gamma(T^+)$. Repeat the procedure until $t = t_{\text{max}}$.
-

The reference point $a_j(t)$ is advanced using (3.27), with the tangential velocity V_j^T chosen to preserve the equal-arclength parametrization (7.1). We refer the reader to [21, 42] for further details.

We use a Fourier spectral discretization in space together with the small-scale decomposition described in [42]. Under this decomposition, the stiff leading-order term in the tangent-angle equation is diagonal in Fourier space and is treated implicitly, while the remaining terms are treated explicitly using a first-order IMEX scheme. We replace the second-order time discretization and preconditioned conjugate-gradient iteration of [42] with this first-order scheme and a direct solve using MATLAB’s backslash operator.

7.3. Event detection and defect-patch construction. In our numerical implementation, we have replaced direct evaluation of the reach condition (3.6) with a computable geometric proxy adapted from the *manifold-death* algorithm of Chirco et al. [10]. The proxy detects the impending overlap of diffuse transition layers at the prescribed microscopic length scale and localizes the corresponding defect region. The event-detection procedure is summarized in Algorithm 7.2.

7.4. SAV scheme for local relaxation. The localized relaxation (5.19) is discretized in space by a finite-volume method on the uniform patch grid, with the Neumann conditions (5.20). The cutoff is applied symmetrically in the discrete dissipation operator, thereby freezing the boundary collar, while the discrete counterpart of (5.19d) enforces exact conservation of the phase mass. Pseudo-time integration uses the second-order SAV–BDF2 scheme described in [37]. Each step requires only linear solves with time-independent matrices and dissipates a modified discrete energy. The iteration is terminated using the discrete counterpart of the stopping criterion (5.23).

8. Numerical simulation of coalescence. To test the sharp–diffuse algorithm, we revisit the symmetric four-circle problem of Zhu–Chen–Hou [42, Fig. 14].

Algorithm 7.2 Approximate reach-defect detection

Step 1. *Detection grid and phase indicator.* Construct the piecewise-linear interface $\Gamma_h(t)$ and a uniform Cartesian grid of spacing h in the tubular neighborhood $\mathcal{N}_{R_{\text{tube}}}(t)$ defined by (3.8), where

$$(7.2) \quad h = \frac{2C_{\text{layer}}\varepsilon}{N_{\text{layer}}}, \quad R = 2C_{\text{layer}}\varepsilon, \quad R_{\text{tube}} = C_{\text{layer}}\varepsilon + R + \sqrt{2}h.$$

Evaluate the signed distance and equilibrium phase field using (2.12) and (2.13), and set $c_h = \text{sgn}(\phi_h)$. We use $N_{\text{layer}} = 8$, $C_{\text{layer}} = 2$, and $h = \varepsilon/2$.

Step 2. *Moment signature.* For each grid point x_i satisfying $|d_h(x_i, t)| \leq C_{\text{layer}}\varepsilon$, define the normalized moment tensor by:

$$(7.3a) \quad \mathbb{T}_{R,h}(x_i) := \frac{h^2}{Z_{R,h}} \sum_{x_\ell \in \mathcal{A}_{R,h}(x_i)} (x_\ell - x_i) \otimes (x_\ell - x_i) c_h(x_\ell, t),$$

$$(7.3b) \quad \mathcal{A}_{R,h}(x_i) := \{x_\ell : ||x_\ell - x_i| - R| \leq \tfrac{h}{2}\},$$

$$(7.3c) \quad Z_{R,h} := \frac{\pi}{4} \left[\left(R + \tfrac{h}{2}\right)^4 - \left(R - \tfrac{h}{2}\right)^4 \right].$$

Let $\lambda_{1,h}(x_i) \geq \lambda_{2,h}(x_i)$ be the eigenvalues of $\mathbb{T}_{R,h}(x_i)$. Classify x_i as *thin* if

$$(7.4) \quad \lambda_{1,h}(x_i) > \delta_{\text{sig}}, \quad \lambda_{2,h}(x_i) < -\delta_{\text{sig}}, \quad \delta_{\text{sig}} = 10^{-1}.$$

Step 3. *Event time and defect localization.* The first time at which the set of thin points is nonempty defines the numerical topological event time. Its connected components provide discrete approximations of the reach-defect sets. Enclose each component in a buffered square satisfying (5.1) and (5.1b); these squares define the defect regions Ω used for the localized relaxation.

The initial interface consists of two circles of radius 1 centered at $(\pm 1.25, 0)$ and two circles of radius 0.9 centered at $(0, \pm 2)$. Under the nonlocal Mullins–Sekerka evolution, phase mass is transferred between components, causing the smaller components to contract while the larger components expand and approach one another. The calculation reported in [42] was halted at $t = 0.75$ because the interface velocities became increasingly large near the coalescence.⁵

We continue the calculation through coalescence using the hybrid sharp–diffuse algorithm. The formulation used in [42] is related to ours by

$$(8.1) \quad \tilde{t} = \frac{m_0\sigma}{4}t, \quad \tilde{\mu}_0 = \frac{2}{\sigma}\mu_0, \quad \tilde{\mu}^f = \frac{2}{\sigma}\mu^f, \quad \tilde{V} = \frac{4}{m_0\sigma}V,$$

where the tilde variables denote those used in [42]. Taking $m_0 = \sigma = 2$ gives $\tilde{t} = t$, $\tilde{\mu}_0 = \mu_0$, and $\tilde{V} = V$, reducing (3.25) exactly to the formulation of Zhu–Chen–Hou.

Each interface is initialized with 32 Lagrangian markers and adaptively refined so that the marker spacing does not exceed 2×10^{-2} . This gives 316 markers on each larger circle and 284 on each smaller circle. The principal discretization parameters

⁵It was observed in [42] that the impending coalescence evolves on a much smaller time scale than the macroscale evolution, foreshadowing the fast time scale (5.7) of the localized mixing dynamics.

are

$$(8.2) \quad h = \frac{1}{128}, \quad \varepsilon = 2h = \frac{1}{64}, \quad \Delta t_{\text{BI}} = 2.5 \times 10^{-3}.$$

Here Δt_{BI} is the maximum boundary-integral time step, which is reduced as necessary to satisfy the interface-motion CFL condition. The localized Cartesian patch contains 40×80 grid cells. The event criterion is triggered at

$$(8.3) \quad T = 0.9075.$$

The sharp-interface evolution is then frozen while the localized phase-field system is advanced in pseudo-time using

$$(8.4) \quad \Delta s = 10^{-4}, \quad s_{\text{max}} = 1.41 \times 10^{-2}.$$

After 141 pseudo-time steps, the discrete relative energy-dissipation rate falls below the tolerance in (5.23). The terminal zero level set defines the outgoing interface, from which the boundary-integral calculation is restarted. Over $0 \leq t \leq 1$, the relative error in the conserved total enclosed area is 0.27%. The complete evolution is shown in Figure 5.

For comparison, we compute the fully diffuse Cahn–Hilliard evolution on the periodic square $[-4, 4]^2$ using the SAV–BDF2 Fourier pseudospectral method of [37]. The domain is discretized on an $N_{\text{grid}} \times N_{\text{grid}}$ uniform grid, with

$$(8.5) \quad h = \frac{8}{N_{\text{grid}}}, \quad \varepsilon = 2h = \frac{16}{N_{\text{grid}}}, \quad N_{\text{grid}} = 512, 1024, 2048.$$

To compare the fully diffuse and hybrid solutions at $t = 1$, we reconstruct a phase field from the final sharp interface using the equilibrium profile (2.13).

As shown in Figure 6, the final morphology depends strongly on ε . Larger values produce earlier coalescence, leaving more time for interfacial relaxation and nonlocal mass transfer to eliminate the smaller components. At $\varepsilon = 1/128$, by contrast, the fully diffuse solution retains both smaller components at $t = 1$, while the central component remains distinctly dumbbell-shaped. The sharp–diffuse method captures this qualitative small- ε behavior at substantially lower cost. The computational runtimes reported in Table 1 show speedup factors of approximately 68 and 386 at $N_{\text{grid}} = 1024$ and 2048, respectively. Meanwhile, at $\varepsilon = 1/256$ and $N_{\text{grid}} = 4096$, the sharp–diffuse method simulates the evolution in just 86 seconds, whereas the corresponding fully diffuse calculation is prohibitively expensive on the current computational platform.

TABLE 1
Wall-clock times for the four-circle coalescence problem.

Method	N_{grid}	ε	T_{CPU}
Diffuse	512	1/32	4 min
	1024	1/64	35 min
	2048	1/128	4.5 h
Sharp–Diffuse	1024	1/64	31 s
	2048	1/128	42 s
	4096	1/256	86 s

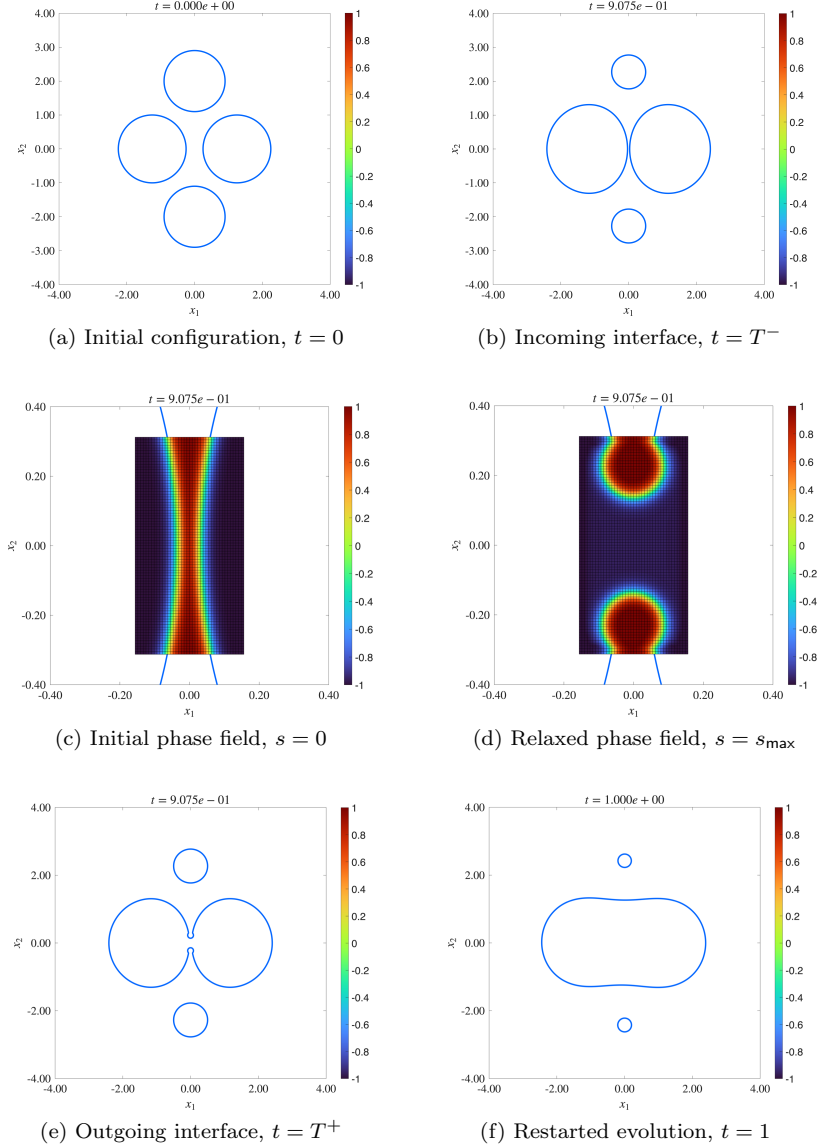


FIG. 5. **Simulation of coalescence using the sharp-diffuse interface model.** The four-circle problem of Zhu–Chen–Hou [42, Fig. 14] is simulated using Algorithm 7.1 with the parameters in (8.2) and (8.4). The initial configuration in panel (a) evolves according to the boundary-integral formulation (Proposition 3.8) of the Mullins–Sekerka system (3.20). While the transition layers remain separated, Theorem 3.4 identifies this system as the formal sharp-interface limit of the Cahn–Hilliard system (2.2). As the two larger components approach one another, the geometric separation required by Assumption 3.2 degenerates. The event detector in Algorithm 7.2 halts the sharp-interface evolution at the event time (8.3), shown in panel (b). The corresponding sharp-interface calculation of Zhu–Chen–Hou was terminated at $t = 0.75$ prior to coalescence, and continued no further. Here, the localized phase-field system (5.19) is instead used to resolve the topological transition, as shown in panels (c)–(d); during this relaxation, the macroscopic time t is frozen, with s denoting pseudo-time. The terminal zero level set defines the outgoing interface in panel (e), from which the sharp-interface evolution is restarted and evolved until the final time $t = 1$, as shown in panel (f). A video of the complete evolution (a)–(f) is available at [35].

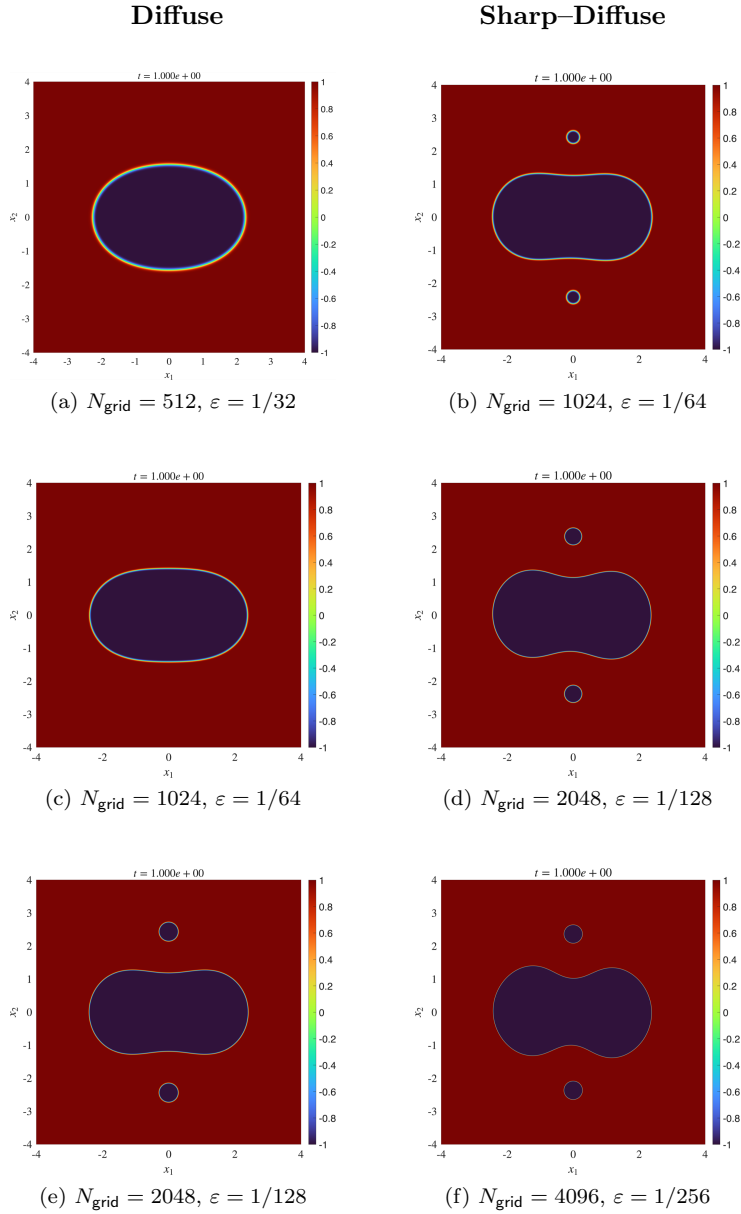


FIG. 6. *Comparison of diffuse and sharp-diffuse simulations of coalescence as $\varepsilon \rightarrow 0$. The four-circle problem of Zhu–Chen–Hou [42, Fig. 14] is simulated using the fully diffuse SAV–BDF2 Fourier pseudospectral method of [37] and the sharp-diffuse interface model in Algorithm 7.1. The phase fields at $t = 1$ are displayed, which show that the final morphology depends strongly on ε . Larger ε produces earlier coalescence, leaving more time for surface tension to round the connected interface and accelerate the disappearance of the smaller components. Decreasing ε delays both processes; in the fully diffuse calculation with $N_{\text{grid}} = 2048$, the smaller components persist at $t = 1$ and the coalesced component retains a dumbbell shape. The hybrid method reproduces this qualitative small- ε behavior at substantially lower cost. As reported in Table 1, at $N_{\text{grid}} = 2048$ it is approximately 386 times faster than the corresponding diffuse calculation.*

Finally, we characterize the immediate post-coalescence dynamics by computing the bridge radius,

$$(8.6) \quad \begin{aligned} r_b(t) &:= \frac{x_2^+(t) - x_2^-(t)}{2}, \\ x_2^+(t) &:= \min\{x_2 \geq 0 : (0, x_2) \in \Gamma(t)\}, \\ x_2^-(t) &:= \max\{x_2 \leq 0 : (0, x_2) \in \Gamma(t)\}. \end{aligned}$$

In studies of liquid-drop coalescence, the bridge radius is a standard measure of the post-contact dynamics. At early times, it typically obeys a self-similar law $r_b \sim t^\alpha$, with the exponent α determined by the geometry and dominant transport mechanism [17, 25]. In particular, the bulk diffusion considered here is predicted to produce the exponent $\alpha = 1/5$ [17, Sec. 2.5], a growth law that has been observed experimentally during the sintering of metal spheres [24].

Specifically, if R is the macroscopic radius of curvature of the incoming interfaces, then the characteristic axial half-width satisfies $w \sim r_b^2/R$ for $r_b \ll R$. Over this axial scale, the interface undergoes a radial displacement of order r_b , so the radius of curvature at the bridge tip satisfies $r_c \sim w^2/r_b \sim r_b^3/R^2$, and hence $\kappa_b \sim R^2/r_b^3$. The Gibbs–Thomson condition (3.20b) then gives the chemical-potential scale $|\mu_0| \sim \sigma R^2/r_b^3$. Assuming that the corresponding harmonic field varies over the radial bridge scale r_b , its normal derivative jump satisfies $|\partial_n \mu_0| \sim |\mu_0|/r_b \sim \sigma R^2/r_b^4$. Substituting into the velocity law (3.20c) and integrating yields the self-similar law

$$(8.7) \quad r_b(t) = [r_0^5 + A(t - T)]^{1/5}, \quad t \geq T,$$

where T is the topological event time, $r_0 = r_b(T^+)$ is the initial post-transition bridge radius, and $A > 0$ is a constant which can be approximated by least-squares fitting over the time interval $[T, t_{\max}]$. We compute the bridge growth for the sharp-diffuse simulations with $N_{\text{grid}} = 1024, 2048$, and 4096 , for which the event times are $T = 0.9075, 0.9522$, and 0.9733 , respectively. The results in Figure 7 show increasingly close agreement with the predicted law (8.7) as the resolution is increased.

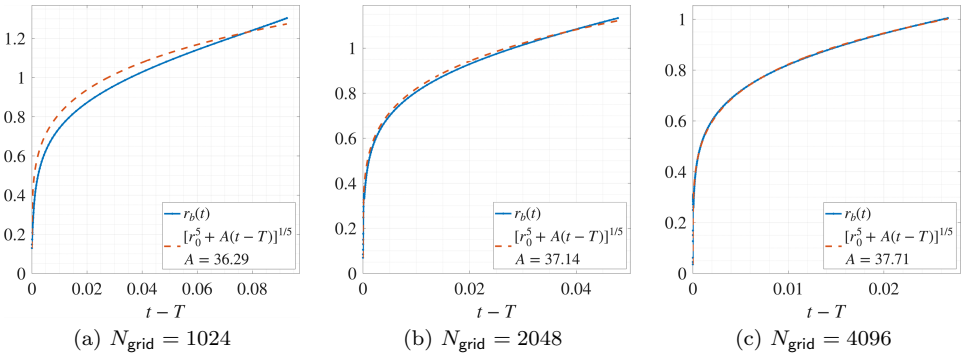


FIG. 7. **Self-similar bridge growth following coalescence.** The bridge radius (8.6) is shown for three sharp-diffuse simulations. The dashed curves are least-squares fits to the predicted law (8.7) over $[T, t_{\max}]$. Agreement with the predicted scaling improves as the resolution is increased.

9. Conclusions. We have developed a sharp-diffuse interface model for intermittent and isolated topological transitions in Cahn–Hilliard phase coarsening. The mathematical justification for the model is based on a geometric characterization of the sharp-interface regime and established analytical results for the limiting Mullins–Sekerka system. The model couples efficient boundary-integral evolution with spacetime-localized diffuse relaxation. Numerical tests demonstrate continuation through coalescence and speedups of two to three orders of magnitude over a fully diffuse solver. In the future, we plan to extend the model to more complex interfacial flows. A preliminary sharp-diffuse computation for a vortex sheet with surface tension is shown in Figure 8, demonstrating continuation through the *pinching* singularities observed by Hou–Lowengrub–Shelley [21, Figure 11].

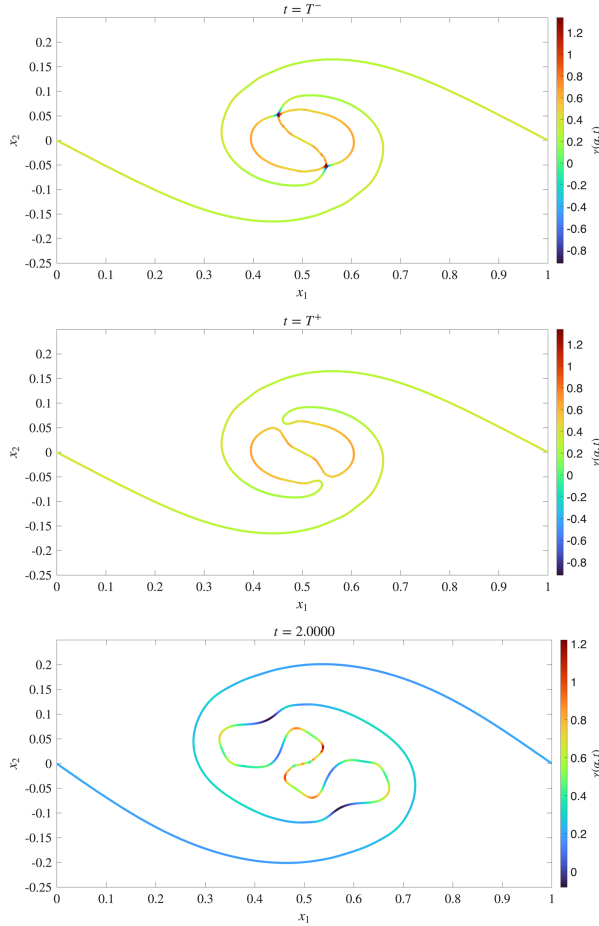


FIG. 8. *Sharp-diffuse continuation through vortex-sheet pinch-off.* We consider the periodic vortex-sheet problem of Hou–Lowengrub–Shelley [21, Figure 11], in which a sinusoidal perturbation of a flat sheet rolls up under the Kelvin–Helmholtz instability in a homogeneous fluid at Weber number $We = 200$. This preliminary computation uses $N = 1024$ interface points and transition-layer width $\varepsilon = 2h = 1/512$, where $h = 1/1024$ is the local grid spacing. The interface is shown at the topological event time $t = T^-$ (top), following localized diffuse relaxation at $t = T^+$ (middle), and after restarting the sharp-interface evolution at the final time $t = 2$ (bottom). The line color denotes the vortex-sheet strength γ . The topological event is detected at $T = 1.5$, whereas the computations in [21] terminated at $t = 1.4$. The symmetry of the configuration produces two simultaneous topological events; the sharp-diffuse solution preserves this symmetry through the topology change.

TABLE 2
Structural analogy between sharp-diffuse continuation and shock development.

	Sharp-Diffuse	Shock development
Regular evolution	Mullins-Sekerka	Smooth compressible Euler
Singularity	Topological transition	Shock formation
Tracked interfaces	Material	Fast acoustic characteristic
Physical breakdown	Overlap of diffuse transition layers	Collapse of acoustic characteristics
Weak continuation	Varifold solution	Convex integration
Strong continuation	Localized Cahn-Hilliard evolution	Free-boundary shock development
Outcome	Smooth interfaces satisfying the separated-layer condition	A persistent shock separating smooth flow regions

Several analytical questions remain open. Although the classical Cahn-Hilliard and Mullins-Sekerka stages rest on established theory, the interface-surgery construction remains formal and is justified only for an idealized coalescence geometry. For perturbations of this geometry, an implicit-function argument could establish both $C^{3+\alpha}$ regularity of the outgoing interface and the required lower bound on its reach. From the numerical point of view, simulating more complex phase-coarsening problems, particularly in three dimensions, will require a robust, parallel implementation with an event-local hierarchical data structure [34].

Finally, we remark that the proposed methodology may be placed in the broader context of methods for continuation through singularities. The current program seeks continuation through topological singularities; an analogous program is development of shock singularities for the multidimensional compressible Euler equations. In both settings, weak solutions do not select a sufficiently resolved continuation, motivating two stages: classical evolution followed by strong continuation. Here, Mullins-Sekerka evolution is followed by localized Cahn-Hilliard dynamics. In the shock development program, classical compressible Euler evolution is followed by free-boundary shock development; see [12, 39] for the analysis of the shock formation stage, and [13, 4] for the shock development stage.

The analogy between the two programs, summarized in Table 2, is architectural: a geometric criterion degenerates, and a second problem constructs the strong continuation. Two differences are essential. First, the present program is algorithmic and partly formal, whereas the cited shock constructions are rigorous. Second, our topological events are intermittent and isolated, after which a smooth interface re-emerges. By contrast, the initial shock formation is followed instantaneously by a continuum of singularities (i.e., shock propagation), while weaker cusp and characteristic singularities also emanate from the first singularity, further complicating the problem.

Appendix A. Polynomial profile for idealized coalescence.

We derive the explicit polynomial height function stated in Example 6.3. For prescribed values of L , H , and $h(0)$, we show that it is the unique even polynomial of degree at most twelve satisfying the attachment and area constraints in Assumption 6.2. We then verify that (6.11) guarantees positivity and a nondegenerate minimum. The polynomial was derived with the assistance of ChatGPT and independently verified by direct substitution.

A.1. Polynomial ansatz. Recall from Example 6.3 the dimensionless parameters Λ and α , the symmetric variable r , and the representation (6.10b). The conditions (6.6) and (6.7) imply that $\Lambda > 1$ and $0 < \alpha < C_{\text{layer}}/H < 1$.

Since r depends only on ξ_1^2 , the representation (6.10b) enforces evenness automatically. Conversely, every even polynomial h of degree at most twelve admits a unique representation of this form with $\mathcal{P}$ a polynomial of degree at most six. The attachment and central-height conditions become

$$(A.1) \quad \mathcal{P}(0) = 1, \quad \mathcal{P}(1) = \alpha.$$

Finally, introducing $s = \xi_1/L$, the area constraint (6.9) becomes

$$(A.2) \quad \int_0^1 \mathcal{P}(1 - s^2) ds = \frac{\pi}{4\Lambda}.$$

A.2. Imposing the attachment conditions. Since $\mathcal{P}$ has degree at most six and $\mathcal{P}(0) = 1$, we may write

$$(A.3) \quad \mathcal{P}(r) = 1 + \sum_{\ell=1}^6 c_\ell r^\ell.$$

The derivative attachment conditions (6.5) and (6.8) determine the first four coefficients. By evenness, it suffices to impose these conditions at $\xi_1 = L$. Rather than differentiate the composite expression $h(\xi_1) = H\mathcal{P}(1 - \xi_1^2/L^2)$ four times, we compare its Taylor expansion at the attachment point. Set

$$(A.4) \quad \delta := \frac{\xi_1 - L}{L}, \quad s = 1 + \delta, \quad r = -2\delta - \delta^2.$$

It follows that

$$(A.5) \quad \mathcal{P}(-2\delta - \delta^2) = \frac{h(L + L\delta)}{H}.$$

Expanding about $\delta = 0$ gives

$$h(L + L\delta) = h(L) + Lh'(L)\delta + \frac{L^2 h''(L)}{2} \delta^2 + \frac{L^3 h'''(L)}{6} \delta^3 + \frac{L^4 h''''(L)}{24} \delta^4 + \mathcal{O}(\delta^5).$$

Using the derivative attachment conditions and $\Lambda = L/H$, we obtain

$$(A.6) \quad \mathcal{P}(-2\delta - \delta^2) = 1 - \frac{\Lambda^2}{2} \delta^2 - \frac{\Lambda^4}{8} \delta^4 + \mathcal{O}(\delta^5).$$

Substituting $r = -2\delta - \delta^2$ into (A.3) and comparing with (A.6) yields

$$(A.7) \quad c_1 = 0, \quad c_2 = -\frac{\Lambda^2}{8}, \quad c_3 = -\frac{\Lambda^2}{16}, \quad c_4 = -\frac{\Lambda^4 + 5\Lambda^2}{128}.$$

The coefficients c_5 and c_6 remain undetermined. For convenience, set

$$(A.8) \quad A := c_5 + c_6, \quad B := -c_6.$$

With this notation, (A.3) takes the form (6.10c).

A.3. Imposing the remaining constraints. It remains to determine A and B . The central-height condition $\mathcal{P}(1) = \alpha$ in (A.1) gives precisely the coefficient A in (6.10d). To impose the area constraint, we use the elementary identities

$$(A.9a) \quad \int_0^1 (1-s^2)^2 ds = \frac{8}{15}, \quad \int_0^1 (1-s^2)^3 ds = \frac{16}{35}, \quad \int_0^1 (1-s^2)^4 ds = \frac{128}{315},$$

$$(A.9b) \quad \int_0^1 (1-s^2)^5 ds = \frac{256}{693}, \quad \int_0^1 s^2(1-s^2)^5 ds = \frac{256}{9009}.$$

Substituting (6.10c) into (A.2) and using (A.9) gives

$$(A.10) \quad \frac{\pi}{4\Lambda} = 1 - \frac{\Lambda^2}{9} - \frac{\Lambda^4}{315} + \frac{256}{693}A + \frac{256}{9009}B.$$

Solving (A.10) for B gives (6.10e). Thus the prescribed central height determines A , after which the area constraint uniquely determines B .

A.4. Positivity and nondegeneracy. Assume the parameter constraints given in (6.11) and define

$$(A.11) \quad D := \frac{\Lambda^4 + 19\Lambda^2}{32} - 5A + B.$$

The second condition in (6.11) gives $D > 0$. Differentiating (6.10b) gives

$$h''(0) = -\frac{2H}{L^2}\mathcal{P}'(1).$$

From (6.10c), we compute $\mathcal{P}'(1) = -D$, and therefore

$$(A.12) \quad h''(0) = \frac{2H}{L^2}D > 0.$$

Thus $D > 0$ gives the required nondegeneracy at the center.

To establish positivity, write $\mathcal{P}$ in the degree-six Bernstein basis:

$$(A.13) \quad \mathcal{P}(r) = \sum_{k=0}^6 \beta_k \binom{6}{k} r^k (1-r)^{6-k}.$$

Conversion from the power-basis representation (6.10c) gives

$$(A.14a) \quad \beta_0 = \beta_1 = 1, \quad \beta_2 = 1 - \frac{\Lambda^2}{120}, \quad \beta_3 = 1 - \frac{9\Lambda^2}{320},$$

$$(A.14b) \quad \beta_4 = 1 - \frac{25\Lambda^2}{384} - \frac{\Lambda^4}{1920}, \quad \beta_5 = \alpha + \frac{D}{6}, \quad \beta_6 = \alpha.$$

The bound $1 < \Lambda < 5/2$ implies $\beta_k > 0$ for $0 \leq k \leq 4$, while $\alpha > 0$ and $D > 0$ imply $\beta_5, \beta_6 > 0$. Since the Bernstein basis functions are nonnegative on $[0, 1]$ and sum to one, (A.13) gives $\mathcal{P}(r) > 0$ for $r \in [0, 1]$. Consequently, $h(\xi_1) > 0$ throughout $[-L, L]$.

Acknowledgments. This work was supported by the Mark Kac Applied Mathematics Postdoctoral Fellowship at the Center for Nonlinear Studies at Los Alamos National Laboratory. OpenAI's ChatGPT was used to assist with the algebraic derivation of the polynomial profile in Section A. The calculation was independently verified by direct substitution, and the author assumes full responsibility for all content of the manuscript.

Los Alamos National Laboratory Report LA-UR-26-27597.

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